

Controlled Source DAS Coupling Tests: Implications for Unburied Deployment on the Moon and Earth

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November 03, 2025

Abstract

Moonquake recordings are key to understanding the Moon's internal structure, yet existing data from the Apollo missions are limited. Distributed Acoustic Sensing (DAS) offers a promising alternative to conventional seismometers for future lunar seismic monitoring, particularly due to its robustness and suitability for high-scattering environments like the lunar regolith. Since burial of DAS cables may be impractical on the Moon, understanding the coupling performance of unburied cables is critical. In this study, we present experimental results from controlled shaker tests in a lunar regolith simulant, comparing the coupling performance of buried and unburied fiber-optic cables with varying mechanical properties and assessing the impact of wind noise and surface undulations. Our findings show that burial generally improves signal amplitude and phase reliability, while unburied cables yield weaker signals and, in some cases, exhibit phase shifts. However, thicker and stiffer unburied cables achieve comparable performance as buried cables, maintaining stable amplitude ratios and phases. Furthermore, we demonstrate that free-hanging cable segments degrade coupling quality substantially for all tested cables, emphasizing the importance of continuous ground contact. We also show that wind is a major contributor to poor signal quality in unburied DAS deployments on Earth - an issue that would not arise on the Moon due to its lack of atmosphere. These results suggest that with appropriate cable selection and deployment strategies, unburied DAS is a viable option for lunar seismic applications.

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Key Points:

- Unburied distributed acoustic sensing can be feasible on the Moon with suitable cable choice and careful deployment
- Thick unburied cables exhibit flat amplitude and phase spectra, performing comparably to buried cables
- Wind and surface undulations degrade signal quality in unburied terrestrial deployments, especially for thin cables

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Abstract

Moonquake recordings are key to understanding the Moon’s internal structure, yet existing data from the Apollo missions are limited. Distributed Acoustic Sensing (DAS) offers a promising alternative to conventional seismometers for future lunar seismic monitoring, particularly due to its robustness and suitability for high-scattering environments like the lunar regolith. Since burial of DAS cables may be impractical on the Moon, understanding the coupling performance of unburied cables is critical. In this study, we present experimental results from controlled shaker tests in a lunar regolith simulant, comparing the coupling performance of buried and unburied fiber-optic cables with varying mechanical properties and assessing the impact of wind noise and surface undulations. Our findings show that burial generally improves signal amplitude and phase reliability, while unburied cables yield weaker signals and, in some cases, exhibit phase shifts. However, thicker and stiffer unburied cables achieve comparable performance as buried cables, maintaining stable amplitude ratios and phases. Furthermore, we demonstrate that free-hanging cable segments degrade coupling quality substantially for all tested cables, emphasizing the importance of continuous ground contact. We also show that wind is a major contributor to poor signal quality in unburied DAS deployments on Earth - an issue that would not arise on the Moon due to its lack of atmosphere. These results suggest that with appropriate cable selection and deployment strategies, unburied DAS is a viable option for lunar seismic applications.

Plain Language Summary

Understanding the Moon’s interior is important for planetary science, and seismic measurements are one of the most effective tools for this purpose. These measurements rely on detecting moonquakes: vibrations that travel through the lunar subsurface. However, the only available moonquake data were collected during the Apollo missions using a small number of traditional seismometers. Distributed Acoustic Sensing (DAS) is a newer technique that could improve future lunar seismic monitoring. DAS uses a fiber-optic cable to measure ground motion along its entire length, effectively turning the cable into thousands of vibration sensors. This method offers dense spatial coverage and could simplify deployment compared to conventional instruments. On Earth, DAS cables are typically buried to ensure good contact with the ground. However, on the Moon, burial may not be feasible. To investigate whether unburied cables can still provide reliable data, we conducted controlled experiments using a lunar soil simulant and several types of fiber-optic cables. We found that thicker, stiffer cables can record stable signals even when unburied, provided they maintain contact with the ground. These results suggest that DAS could be a viable method for lunar seismic sensing, enabling large-scale measurements without the need for complex installation procedures.

1 Introduction

Currently, the only seismic dataset recorded on the Moon originates from the Apollo missions, during which both active and passive seismic experiments were conducted. Five stations deployed on the near side of the Moon recorded over 12,000 seismic events from 1969-1977 (see Nunn et al. (2020) for a review of the instrumentation and collected data). This dataset has led to the identification of four distinct types of natural seismic sources on the Moon, namely, deep moonquakes, shallow moonquakes, thermal moonquakes, and meteoroid impacts (Nakamura et al., 1982). The collected seismic data have been the primary source for estimating the lunar seismic velocity structure and have been extensively analyzed over the decades and continue to receive fresh scrutiny, for example, by Sollberger et al. (2016), who used spatial wavefield gradients to construct an elastic model of the shallow lunar subsurface. A major challenge in interpreting moonquake seismograms is the presence of long-duration and high-amplitude coda waves that are much more

pronounced than exhibited in earthquake seismograms. Caused by the strongly scattering and low attenuating lunar shallow subsurface, the coda waves obscure later arriving seismic phases.

An approach that could help address this challenge is an emerging sensing technology called Distributed Acoustic Sensing (DAS), which measures dynamic strain over long distances. DAS has rapidly gained traction in a broad range of geophysical applications as an alternative to conventional ground motion sensors such as geophones and seismometers. A DAS setup consists of a fiber-optic cable connected to an interrogator. The interrogator sends laser pulses into the optical fiber and measures the returning light that experiences Rayleigh back-scattering from naturally occurring inhomogeneities in the fiber smaller than the probing wavelength. This backscattered light undergoes phase changes as the scatterers move due to strain on the fiber, consequently, the dynamic strain of the optical fiber can be inferred, thereby transforming the optical fiber cable into a dense array of vibration sensors. As a result, DAS is sensitive to a length change of the cable which - in the case of perfect coupling of cable to the ground - corresponds to the inline strain.

DAS is a technology that is robust and easy to deploy, making it a promising candidate for lunar deployment. A single fiber-optic cable attached to an interrogator can record over tens of kilometers and replace thousands of sensors, eliminating the need for individual power supplies or data acquisition systems. Fiber-optic sensors can withstand harsh environmental conditions, including extreme temperatures and radiation. They have been successfully deployed in environments ranging from cryogenic temperatures (Yang et al., 2021) to temperatures as high as 1000°C in ϕ -OTDR-based DAS experiments (Xu et al., 2025). J. Wu et al. (2021) demonstrated that OFDR-based fiber sensors maintained stable Rayleigh backscattering profiles over multiple months inside a nuclear reactor core, exposed to both high temperatures and neutron radiation. Additionally, long-duration space exposure experiments have shown that fiber-optic cables can remain functional after years in orbit, even when unshielded (e.g., Hartmayer et al., 1993). This study confirms that cables can endure temperature cycling, radiation exposure, and polymer aging in space. However, planned mission duration is critical since radiation effects and material degradation accumulate over time, and the small but non-negligible risk of micrometeoroid impacts on the cable and a resulting fiber breakage increases with exposure time. Recent modeling studies have demonstrated the potential of DAS for lunar applications (W. Wu et al., 2024). The dense channel spacing that DAS offers is particularly suited for high-scattering environments such as the lunar shallow subsurface and could allow DAS to recover deep seismic phases that were obscured in Apollo data due to scattering. Zhai et al. (2024) estimate that DAS could detect around 60 % of the moonquakes observed by the Apollo seismometers. However, with foreseeable improvements in DAS technology this number could rise to detect nearly all moonquakes previously observed by Apollo. Their findings emphasize the critical role of DAS performance in moonquake detectability, underscoring the relevance of understanding strain transfer and amplitude behavior as investigated in this study.

However, several open questions need to be addressed before DAS can be deployed on the Moon. A key challenge is determining how to best deploy the fiber-optic cable ensuring good coupling. On Earth, burial is commonly used to ensure good coupling, and studies have shown that it significantly improves signal quality compared to unburied deployment (Dou et al., 2017; Rossi et al., 2022). However, burial may be infeasible in the lunar environment. Unburied deployments on Earth often exhibit poor signal quality, which has been partly attributed to atmospheric acoustic noise, particularly wind and thunder (Carr et al., 2025; Viens et al., 2025; Hudson et al., 2021). This atmospheric noise is not expected to be a significant source of noise, as the Moon has only a thin exosphere. However, it is still unclear whether the poor performance of unburied fiber is primarily due to poor ground coupling or rather due to atmospheric noise. Understanding this distinction is critical for assessing whether unburied deployment on the Moon can yield satisfactory seismic data.

This study employs lunar regolith simulant (LRS) to examine the relationship between the conditions of a fiber optic cable deployment and data quality, highlighting implications for lunar exploration. We are building upon the work of Zandanel et al. (n.d.) which used the same experimental setup to record local and regional earthquakes and found that unburied fiber is capable of detecting earthquakes with a weaker amplitude than buried fiber, but with only a modest decrease in the signal-to-noise. This study aims to quantify this effect and investigate whether the observed amplitude attenuation factor is frequency-dependent and if the phase recorded with unburied fiber is reliable. Using a shaker source to produce controlled and repeatable ground motion in the LRS, we examined the coupling of different cable types with varying mechanical properties, such as outer diameter and weight. Employing cables with drastically different properties is motivated by two key considerations: (1) comparing the performance of cables with differing characteristics can help elucidate the physical mechanisms that govern fiber-regolith coupling and how these mechanisms depend on cable properties, and (2) for lunar deployment, weight and robustness are critical factors. Due to the high cost of transport to the Moon, lightweight and robust cables are highly desirable, especially given that deployment strategies are still under development, with potential concepts of launching the cable onto the surface. Additionally, we investigated the impact of ambient wind to understand how much wind noise contributes to the poor signal quality of unburied fiber observed on Earth. Lastly, we investigated the impact of small-scale surface topography on coupling quality to assess the robustness of unburied cables to surface heterogeneities.

2 Methods

2.1 Experimental setup

An experiment was built that mimics some of the conditions that are present on the Moon. A 9.5-meter-long trough was filled with a generic lunar regolith simulant (LRS), composed of crushed basalt (Figure 1c). The trough was filled to a depth of approximately 59 cm with the LRS, forming a flat surface. A detailed characterization of the LRS is provided in Zandanel et al. (n.d.).

Systematically investigating the coupling behavior of various cable types and deployment configurations could only be achieved by exciting the LRS with controlled frequencies and amplitudes. To generate such controlled ground motion, a shaker (Brüel & Kjær 4809) was mounted at the short end of the trough, aligned with the LRS surface, and oriented to excite motion along the longitudinal axis (see Figure 1d). The shaker was driven with a sequence of single-frequency signals ranging from 3 to 30 Hz in 1 Hz increments. Each frequency was excited for 250 periods and repeated at 10 different amplitude levels. Frequencies below 3 Hz were excluded due to shaker limitations, and frequencies above 30 Hz were excluded to avoid the resonance of the trough, which had the first mode at approximately 30 Hz (see Figure 1b). The energy of most natural moonquakes is below 20 Hz (Civilini et al., 2023).

In addition, one experiment simulated an earthquake by driving the shaker with a real earthquake recording as input. We collected DAS data with three cables that exhibit strongly differing mechanical properties: a tight buffered cable with an outer diameter (OD) of 5.5 mm (referred to as OD5.5), a tight buffered cable with an OD of 3 mm (OD3), and a jacketed cable with an OD of 0.9 mm (OD0.9). The properties of those cables are summarized in Table 1.

To investigate the impact of burial on coupling, we deployed the cables at three depths in the LRS: unburied (laid directly on the surface without additional coupling), shallow-buried (at a depth of approximately 2 to 3 cm below the LRS surface), deep-buried (at a depth of approximately 11 cm) (see Figure 1a). Each cable type was deployed in both unburied and shallow-buried configurations. The OD0.9 was deployed in two separate lengths for each of these configurations, providing two independent measurements per depth. The OD5.5, which contains two optical fibers within a single cable, was interro-

gated on both fibers, yielding two measurements from the same physical cable. Only the OD3 was additionally deployed in the deep-buried configuration. To test the robustness of the results and assess the sensitivity of coupling to small variations in deployment, all unburied cables were redeployed after one round of data acquisition. This involved removing and repositioning the fibers on the surface, followed by a repeated acquisition under the same conditions.

DAS data were acquired using a Terra15 Treble+ interrogator at a sampling rate of 3 kHz, with a gauge length and pulse length of 2.45 meters and a channel spacing of 0.82 meters. This configuration was chosen to balance spatial resolution and signal-to-noise ratio. For comparative analysis of the cables and deployment types, we used the central DAS channels at the midpoint of the trough.

A seismometer (Trillium compact 120s) served as a ground-truth reference and was placed at the center of the trough, aligned with the DAS channels and sampled at 250 Hz.

The trough was located indoors, shielding it from natural wind. To simulate wind noise, a fan was positioned approximately 30 cm above the cables at the center of the trough, blowing vertically downward. The fan operated at three intensity settings, corresponding to maximum wind speeds of 2, 2.4, and 3 m/s at a distance of 30 cm away from the fan. Those wind speeds, classified as a "light breeze" on the Beaufort wind scale, are typical of outdoor conditions on Earth.

To assess the impact of surface irregularities, we introduced artificial undulations in the trough by forming grain piles of varying spacings (15-25 cm, 30-50 cm, 60-100 cm). Cables were laid across these piles such that they contacted the ground only at the peaks and were suspended in between (see Figure 1e). The shaker experiment was repeated for each undulation configuration.

Table 1. Specifications of the used fiber-optic cables. OD = outer diameter. SM = single-mode fiber. MM = multi-mode fiber. Cable OD5.5 contains both single-mode and multi-mode fibers, but only the single-mode fibers were interrogated for the measurements.

Cable type	Abbrev.	OD (mm)	Nr. of cores	Buffering	Weight
PS-2S2M-1PU055-01	OD5.5	5.5	2 SM, 2 MM	tight buffered	27 g/m
SMF-28-100	OD0.9	0.9	1 SM	yellow Hytrel tubing	3.1 g/m
Silixa	OD3	3	1 SM	tight buffered	8.6 g/m

2.2 Data processing

To compare the fiber segments, we analyzed the DAS signal recorded at the central position of each segment. A tap test was performed at the midpoint of the trough to identify the central positions, which were localized with an estimated accuracy of half the channel distance. Subsequently, the signal at the central position was interpolated by applying a weighted average of the two adjacent channels. The amplitude response of each fiber segment was characterized as a function of both excitation frequency and amplitude of the shaker. For each excitation frequency, the corresponding time window was divided into 1-second segments. The frequency spectrum of each segment was computed and averaged to obtain a representative spectrum. This approach ensured consistent spectral resolution and comparability across all excitation frequencies. From each averaged spectrum, the amplitude and phase at the excited frequency were extracted. The seismometer data were processed using the same spectral analysis workflow as the DAS data.

This procedure allowed us to investigate how the recorded amplitude, and thus the coupling efficiency between the fiber and the ground, varies with cable type, deployment depth,

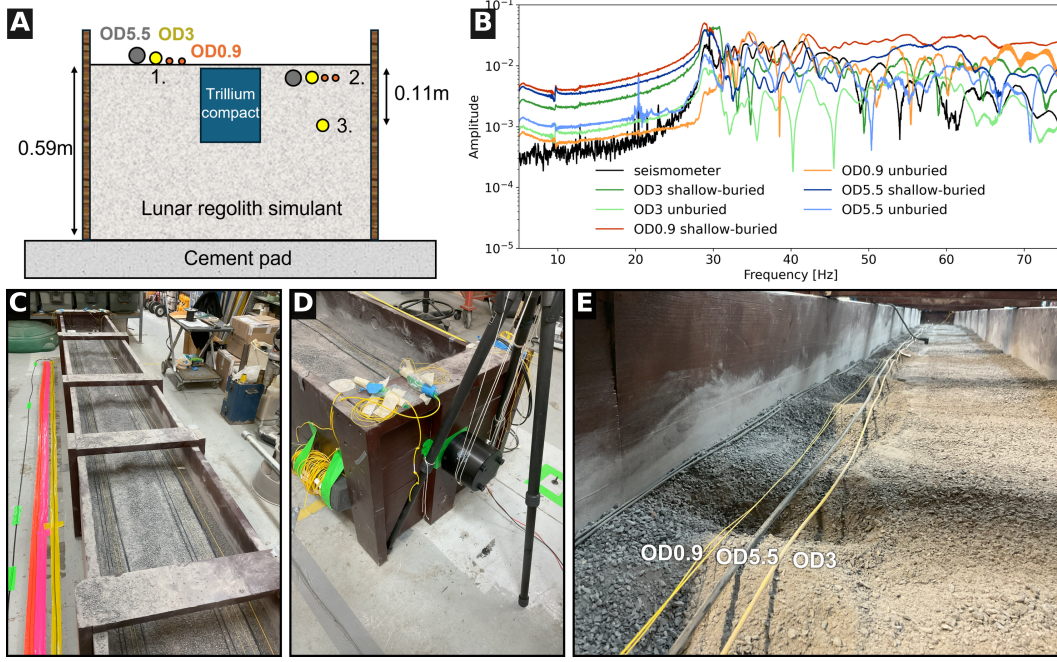


Figure 1. Overview of the experimental setup and conditions. (A) Schematic cross-section of the trough showing the three deployment depths of the fiber-optic cables 1) unburied, 2) shallow-buried, and 3) deep-buried. (B) Amplitude spectrum of a shaker sweep recorded on different fiber segments with minimal undulations, with excitation ranging from 5 to 150 Hz. Resonance behavior present at frequencies above 30 Hz. For comparison, the amplitude spectrum of the co-located seismometer is also shown (normalized for visual clarity), illustrating that the resonance is a property of the experimental trough rather than the fiber coupling. (C) Photograph of the 10-meter-long trough filled with lunar regolith simulant (LRS) and deployed cables. (D) Shaker mounted at the short end of the trough for controlled excitation. (E) Surface undulations created by grain piles (spaced 15-25 cm apart), with cables suspended between contact points to simulate irregular topography.

strain amplitude, and frequency. The amplitudes were divided by the amplitudes measured by the seismometer to account for frequency-dependent variations in shaker efficiency, yielding amplitude ratios. To summarize the coupling performance of each cable and deployment configuration with a single metric, we computed an average amplitude factor. This factor was obtained by averaging the amplitude ratios across all excitation frequencies and amplitudes. Additionally, for the OD5.5 and OD0.9 where two fibers of each cable could be interrogated at the same point - either from two fibers within the same cable (OD5.5) or from two separate cable lengths (OD0.9) - the results from both measurements were averaged to obtain one representative value. An exception is OD0.9 unburied for which the results from both fibers were processed separately, since their performance was very variable, as discussed in more detail later on. The uncertainty of the amplitude response was estimated by accounting for the localization uncertainty of the central channel. Specifically, the analysis was repeated after shifting the central position by ± 0.5 channel spacing, and the resulting variation in amplitude was used to quantify the uncertainty. This channel location uncertainty was considered the dominant source of error, as the shaker was mounted at one end of the trough, resulting in

a strong amplitude gradient along the trough axis. Consequently, even small deviations in channel position could lead to significant differences in recorded amplitude. For the analysis of wind-induced noise, we also focused on the DAS signals recorded at the midpoint of the trough where the fan was positioned. The wind noise data were processed by calculating the power spectral density (PSD) of each trace using Welch's method, with a window length of approximately 3 seconds. This allows us to assess the wind's impact across different frequency bands. To quantify the average intensity of wind-induced vibrations in the seismic frequency band, each trace was further filtered with a fourth-order Butterworth low-pass filter with a cutoff frequency of 20 Hz. The mean absolute amplitude of the filtered signal was then computed in the time domain, providing a representative measure of the sub-20 Hz strain rate noise for each cable segment and wind condition. Here again, unburied cables were redeployed after one round of data acquisition.

3 Results

3.1 Comparison of cable types and deployment

Figure 2a and 2b illustrate how each cable and deployment type responded to shaker excitation, in terms of amplitude and phase. To assess DAS coupling performance, we compare the recorded amplitudes to those of a colocated seismometer, which serves as a ground-truth reference. The seismometer is assumed to have a flat amplitude and phase response across the tested frequency range. Literature suggests that DAS also exhibits flat amplitude and phase spectra when well coupled (Paitz et al., 2021), and this assumption can be verified through comparison with the seismometer. It is important to note that amplitude ratios are not expected to be unity even under ideal coupling conditions, because DAS measures strain rate while the seismometer measures ground velocity. Theoretically, conversion approaches exist to convert DAS recordings to velocity (e.g., Trabattini et al., 2023). However, such conversions assume ideal coupling and long linear cable segments - conditions not met in our experiment. Therefore, we refrain from converting DAS and the seismometer to the same units and instead perform a qualitative comparison using amplitude ratios. The absolute maximum ground particle velocities recorded by the seismometer were on the order of $1\text{e-}4$ m/s. Illustration of the maximum velocities across frequencies and shaker amplitudes is included in the supplementary material.

Shallow and deep-buried fiber segments exhibit relatively constant, frequency-independent amplitude responses. Their phase responses are flat and consistent with the seismometer, indicating reliable strain measurements in the tested frequency-range aside from a uniform amplitude scaling. Unburied fiber segments OD5.5 and OD3 also exhibit stable amplitude factors and flat phases with the exception of an observed phase shift at 25 Hz for OD5.5. Comparison of the two deployed unburied segments of OD0.9 and between repeated deployments OD0.9 shows relatively flat phase and amplitude behavior in some cases, while under less favorable conditions, it exhibits erratic amplitude and phase behavior. In Figure 2, both OD0.9 unburied fiber segments are shown separately to highlight the variability observed between deployments. Repeatability and deployment sensitivity are further discussed in Section 4.1. For one of the unburied OD0.9 segments, a pronounced phase shift is observed at frequencies above 13 Hz, which may indicate that the fiber is resonating at those frequencies.

A comparison across different shaker-driving amplitudes shows a linear relationship between input and recorded amplitude, with no evidence of nonlinear behavior (see supplemental material). To summarize the coupling performance of each cable and deployment configuration with a single metric, we computed an average amplitude response, shown in Figure 2c. For buried cables, the values are relatively similar, with OD0.9 showing the highest amplitude, but still close to the uncertainty range of the other cable types.

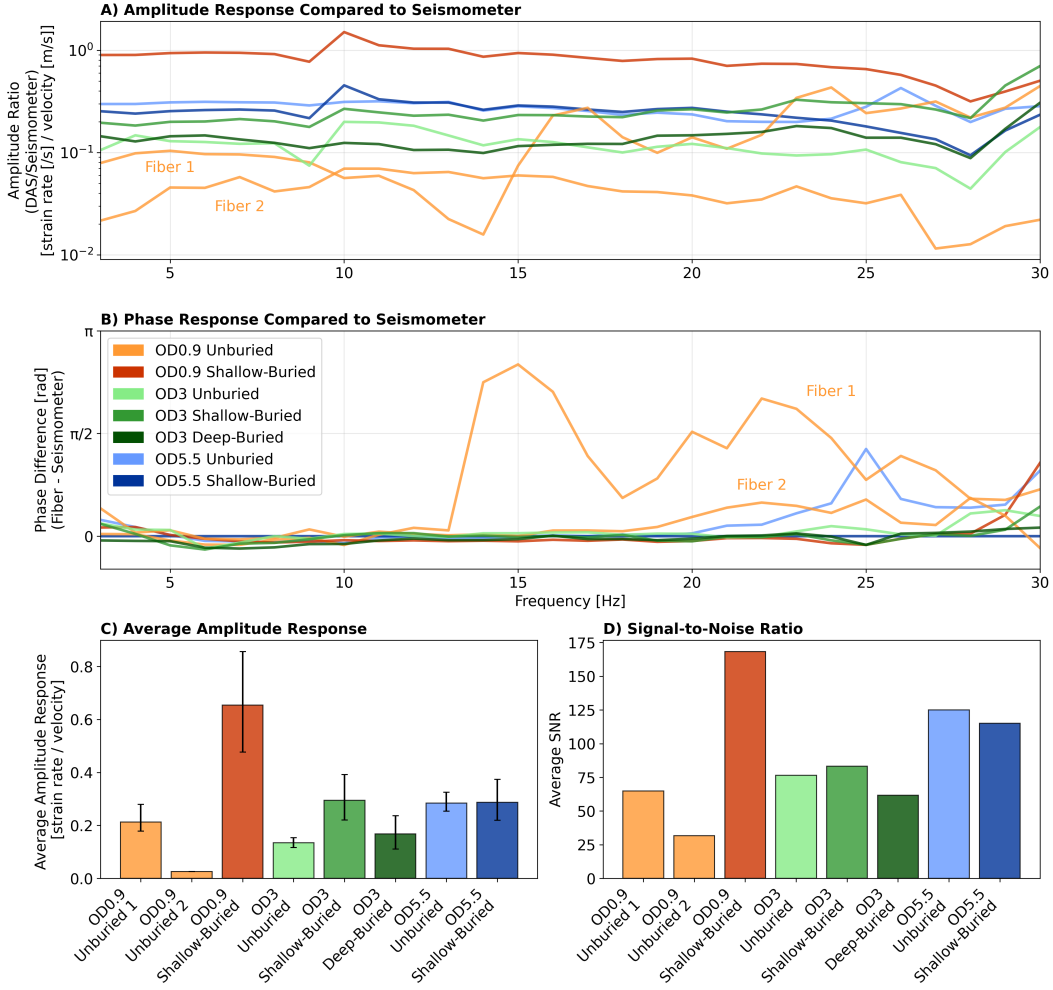


Figure 2. Fiber-optic cable coupling efficiency for different cable types and deployment configurations. (A) Amplitude response showing the ratio of DAS amplitude compared to seismometer amplitude across frequencies 3-30 Hz for shaker amplitude level 7. (B) Phase response showing phase differences of fiber segments relative to the seismometer. (C) Coupling efficiency summary showing amplitude factors averaged across all excited frequencies and amplitudes, with error bars indicating variability due to channel location uncertainty. (D) Signal-to-noise ratio of fiber segments averaged over all excited frequencies and amplitudes. For the OD0.9 unburied configuration, both deployed fibers are shown in all subfigures to illustrate the observed variability between deployments. In contrast, other cables with multiple recordings showed close agreement and are represented by a single curve or value.

In contrast, unburied cables generally exhibit lower amplitude (roughly half that of buried cables), except for OD5.5, which shows comparable values in both configurations. In addition to amplitude and phase analysis, we computed the signal-to-noise ratio (SNR) for each fiber segment, shown in Figure 2d. The SNR was calculated by comparing the signal amplitude during excitation to the noise amplitudes. The SNR trends largely mirror the average amplitude responses, but with some notable differences. OD5.5 consistently shows higher SNR than OD3 in both buried and unburied configurations. Unburied

OD3 achieves SNR values comparable to its buried counterpart, while OD0.9 exhibits significantly lower SNR when unburied.

3.2 Impact of surface undulations

In field deployments, ensuring that fiber-optic cables lie flat and maintain continuous ground contact is labor-intensive. Faster, less controlled deployments may result in cables spanning small surface undulations, potentially degrading coupling quality. To investigate this effect, we introduced artificial surface undulations of varying sizes in the trough. In these deployments, the cables are only in contact with the LRS at the peaks of the undulations. Figure 3 illustrates the impact of these undulations on the signal quality. Coupling quality decreases significantly when cables are suspended across undulations rather than in continuous contact with the ground. For small undulations, the OD0.9 already shows a marked reduction in signal quality across all tested frequencies. In contrast, the OD5.5 and the OD3 maintain relatively good coupling at frequencies below 10 Hz but also exhibit poor performance at higher frequencies. For larger undulations, all unburied cables show substantial signal degradation, characterized by low amplitudes and strong variability. Additionally, phase responses become erratic, with large and inconsistent phase shifts. These drastic phase shifts, especially at higher frequencies may indicate fiber resonance. Notably, the onset frequency of these phase shifts decreases as the undulation spacing increases, suggesting that longer free-hanging segments lower the resonance frequency.

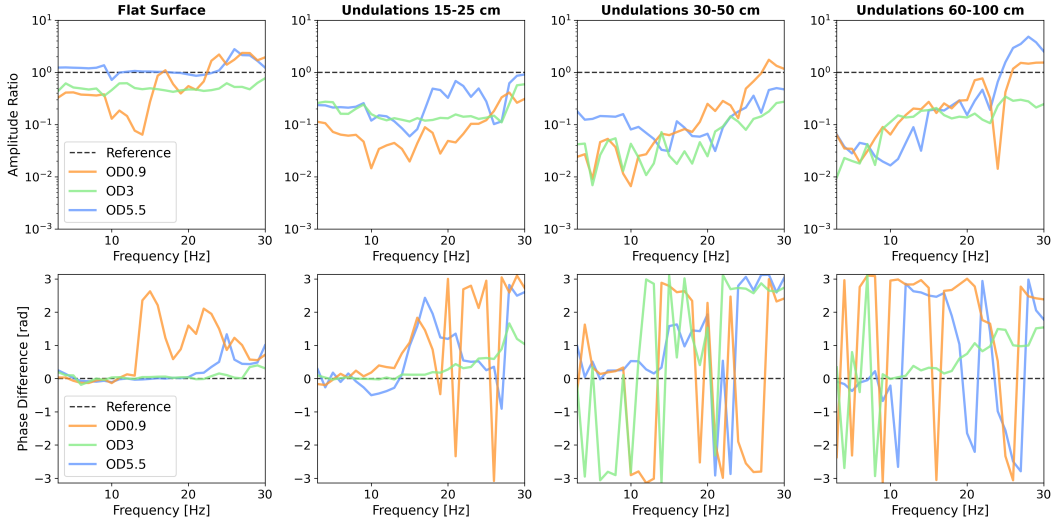


Figure 3. Impact of surface undulations on the coupling of unburied cables (shaker amplitude level 7). Top row: amplitude response plots; bottom row: phase response plots. The leftmost column represents the flat (reference) case (as in Figure 2a and 2b), while the subsequent columns show increasing undulation spacing. All plots compare unburied cables to the same reference: shallow-buried OD5.5. As undulation spacing increases, coupling quality progressively degrades, evidenced by reduced amplitudes and phase shifts, particularly in the two cases with the largest undulations.

3.3 Impact of wind noise

To assess the influence of wind on DAS signal quality, we compare the noise floor of fiber segments exposed to artificially generated wind with those without it. Since the

trough is located indoors, the fibers are naturally protected from ambient wind, allowing for controlled testing. Wind noise might contribute significantly to the poor signal quality observed in unburied cables deployed outdoors. To quantify the contribution of wind on the poor signal quality observed with unburied fiber, we used a fan to simulate wind and analyze the resulting increase in noise levels. The results are presented in Figure 4 (for improved visibility, the spectra are only shown up to 200 Hz). The shallow-buried fibers are practically unaffected by wind, confirming their robustness against wind noise. In contrast, all unburied fibers exhibit elevated noise levels, though the degree of impact varies significantly between cable types. OD5.5 is the least affected, with wind-induced noise primarily below 200 Hz and maximum amplitudes of the wind noise reaching approximately -130 dB/Hz. OD3 shows increased sensitivity, with noise extending up to 400 Hz and peak amplitudes around -125 dB/Hz. OD0.9 is the most affected, with wind noise up to 700 Hz. Particularly strong wind amplitudes are observed in bands between 20-30 Hz and 400-600 Hz, with maximum amplitudes reaching -110 dB/Hz. For seismological applications, the frequency band below 20 Hz is of particular relevance. In this band, the wind-induced noise amplitudes at fan speed 3 are $1.99\text{e-}07$ m/m/s for OD5.5, $7.15\text{e-}07$ m/m/s for OD3, and $2.57\text{e-}06$ m/m/s for OD0.9, sufficient to render moderate earthquakes undetectable.

To illustrate the impact of wind noise on seismic signal detectability, we recorded a shaker-generated simulated earthquake with and without wind noise from the fan. As shown in Figure 5, the presence of wind noise significantly affects the signal detectability. On OD0.9 and OD3 unburied, the event becomes undetectable while on OD5.5 unburied, the signal-to-noise ratio is noticeably reduced. The PSDs of the recordings show similar noise patterns as Figure 4 and are included in the supplementary material.

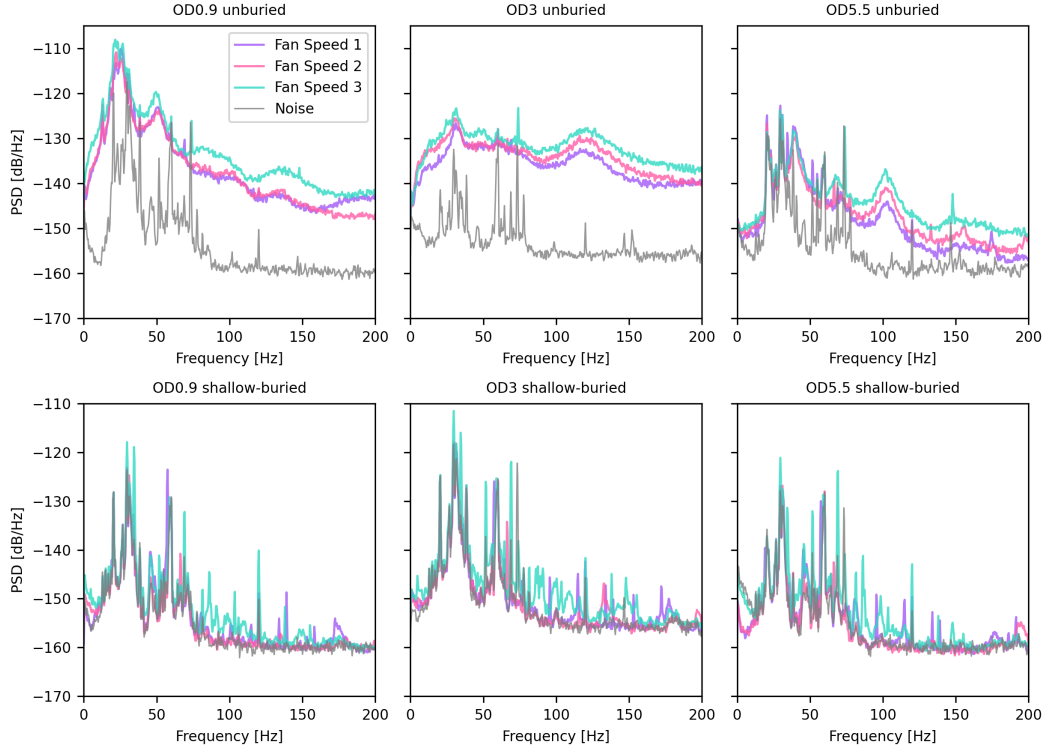


Figure 4. Power spectral density (PSD) comparison of wind-induced noise across different cable segments. Top row: unburied cables (OD0.9, OD3, OD5.5); bottom row: corresponding shallow-buried segments. Each plot shows PSD curves from 0-200 Hz for the four conditions. Fan speeds 1 to 3 correspond to the fan blowing directly onto the cables at increasing wind settings, while the "Noise" condition was recorded with the fan operating but directed away from the cables. This setup captures the ambient noise and fan motor noise without wind impact on the fibers. The difference between the "Noise" and fan speed curves isolates the effect of wind on signal quality.

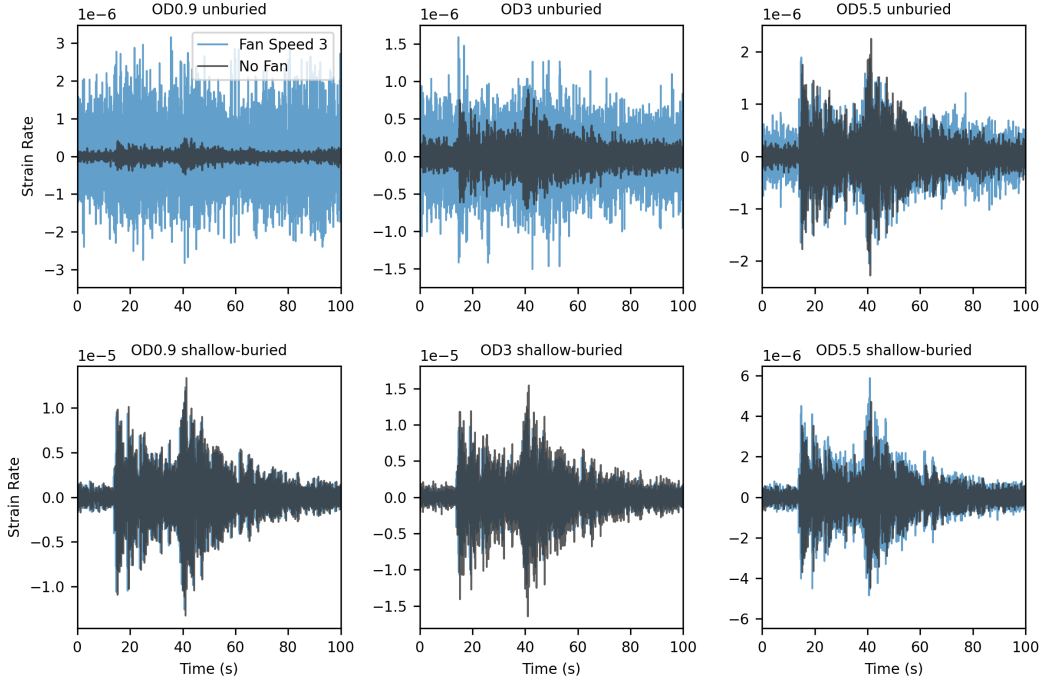


Figure 5. Comparison of simulated earthquake signals recorded on different fiber segments with and without wind noise. The signals were generated with the shaker to simulate a seismic event. Black lines show recordings in the absence of wind, while blue lines show recordings with the fan operating at fan speed 3. All traces were bandpass filtered between 3 and 15 Hz. The simulated earthquake is clearly visible in all traces without wind, although with lower signal strength on OD0.9 unburied and OD3 unburied. With wind present, the seismic signal becomes completely obscured in the OD0.9 and OD3 unburied traces, while it remains visible but with reduced SNR on the OD5.5 unburied trace. All shallow-buried fiber segments show minimal effect from wind noise.

4 Discussion

4.1 Discussion of results

Buried cables consistently recorded signals with a stable amplitude factor and correct phase. Differences of the amplitude factor between cable types were within a factor of two. The deep-buried cable showed slightly lower amplitudes than the shallow-buried ones, but the difference is within the uncertainty range. Since only one cable was tested in the deep-buried configuration, no definitive conclusions can be drawn about the effect of burial depth.

Cable type had a more pronounced impact in unburied deployments. Among the three tested cables, the two thicker ones (OD5.5 and OD3) recorded amplitudes comparable to their buried counterparts and maintained reliable phase information across most frequencies. In contrast, the thinner OD0.9 cable exhibited significantly lower amplitudes and high variability depending on deployment details. Some OD0.9 deployments still yielded correct phase data, albeit with low amplitudes, while others produced highly variable amplitudes and phase disparity. After redeploying the unburied cables for the repeated test, the OD0.9's signal quality degraded noticeably. In the initial deployment, the cables had been in place for some time before data acquisition, allowing them to settle slightly into the fine surface dust and conform more closely to the ground, which likely improved coupling. After redeployment, the cables did not follow the ground as closely, often being slightly lifted by their own stiffness, resulting in reduced contact and poorer coupling. These observations suggest that thin cables are particularly sensitive to small variations in deployment geometry and surface conformity.

The SNR analysis supports these findings. For OD5.5 and OD3, SNR values were similar between buried and unburied deployments, indicating that signal detectability remains robust even without burial. In contrast, OD0.9 showed a substantial reduction in SNR when unburied, consistent with its lower amplitude and higher variability. These results align with the findings of Zandanel et al. (n.d.), who observed reduced amplitudes but comparable SNR for unburied OD3 compared to buried OD3 in earthquake recordings. It is important to note that the noise in our experiment is predominantly environmental and not purely instrumental: Since the noise itself is influenced by the same coupling mechanisms as the signal, the SNR trends partly reflect signal-to-signal comparisons rather than signal-to-instrumental-noise ratios. In a quieter environment like the lunar surface, where environmental noise is absent, the relative SNR differences between cable types may be more pronounced.

Continuous ground contact is critical for signal quality in unburied deployments. Surface undulations that cause cables to hang freely degrade signal quality, leading to unreliable amplitudes and phases. Even short free-hanging segments (~ 20 cm) negatively affected all cable types, with the impact increasing with undulation length. A plausible explanation is that these segments resonate when excited, with resonance frequencies depending on cable properties and undulation length. This interpretation is supported by the observed drastic phase shifts in the presence of undulations. Additionally, some resonance-like behavior was already observed for the unburied OD0.9 even in the absence of undulations, suggesting that this thinner cable is particularly prone to resonating when not well coupled. Resonance effects in DAS cables have also been reported in previous studies, such as Hudson et al. (2021), who observed wind-induced cable resonance in unburied deployments.

Deployment technique significantly influences signal quality. Pressing cables down to ensure ground contact reduces free-hanging segments and improves coupling. This aligns with findings by Mjehovich et al. (2023), who observed improved signal quality when cables were pressed into snow compared to deployment without further coupling efforts, and by Harmon et al. (2022), who reported poor coupling on grassy surfaces where cables were suspended above the ground by the grass.

We found that wind noise significantly increases the noise floor for unburied cables. The OD0.9 was most affected, followed by the OD3 and then the OD5.5. This trend corre-

lates with cable diameter and mass: the lighter OD0.9 was more susceptible to wind-induced motion, which was visibly observed during testing, unlike the heavier OD5.5. However, all cable types exhibited wind noise amplitudes of over $1\text{e-}07$ m/m/s in the 0-20 Hz frequency band, strong enough to mask signals from moderate earthquakes. We illustrated the strong impact of wind noise by comparing recordings of a simulated earthquake with and without artificial wind. A signal that was detectable on all unburied fiber segments in the absence of wind became undetectable on OD0.9 and OD3 unburied in the presence of wind. This suggests that wind is likely a major contributor to the poor signal quality often observed in unburied deployments on Earth, consistent with other studies that have identified wind as a dominant noise source in surface DAS deployments (Carr et al., 2025; Hudson et al., 2021; Viens et al., 2025).

While OD5.5 is representative of the fiber types commonly used in terrestrial DAS applications, we included the much lighter OD0.9 cable in our study to explore options more suitable for lunar deployment, where minimizing weight is a critical factor. Although cables like OD0.9 would not be typically used in terrestrial surface DAS applications, understanding its coupling behavior helps assess the feasibility of lightweight DAS systems for future lunar seismic monitoring. The experimental results for OD5.5 provide insight into the dominant factors affecting signal quality in unburied terrestrial deployments. This cable maintained stable amplitude and phase responses even when unburied throughout the tested frequency range, with only minor degradation compared to its shallow-buried configuration. Wind did lead to a significant increase of the noise floor, but the cable remained capable of detecting strong seismic events. In contrast, surface undulations had a strong detrimental impact: already relatively short (15-25 cm) free-hanging segments led to significant amplitude reduction and erratic phase behavior, particularly at higher frequencies. These results suggest that unburied DAS can achieve signal quality comparable to buried configurations in the absence of strong wind and if continuous ground contact is maintained. The feasibility of this depends on the deployment environment, but strategies such as pressing the cable down or flattening the surface could be employed to increase cable-ground contact.

4.2 Coupling mechanisms

Our experimental results show that signal quality in unburied fiber deployments varies strongly with cable type and deployment details. Thinner cables and configurations with reduced ground contact consistently yielded lower amplitudes and less reliable phase information. While these empirical patterns are clear, the physical mechanisms that govern strain transfer from the ground to the optical fiber remain poorly understood. In this section, we discuss possible coupling mechanisms and present a hypothesis that may explain the observed variability in signal quality. We begin by outlining the two-step process of strain transfer from the ground to the fiber and then explore how deployment geometry and cable properties influence each step. Strain transfer from the ground to the optical fiber involves two steps: 1) Ground-to-cable-coupling: Transfer of strain from the ground to the cable's outer surface; 2) Cable-to-fiber transfer: Transmission of strain through the cable layers to the optical fiber. The second step depends only on the cable's internal mechanical properties. Reinsch et al. (2017) developed a theoretical model for strain transfer in multilayer fiber-optic cables, showing that the efficiency of strain transmission from the cable surface to the optical fiber depends on cable geometry and material properties, and varies with wavelength. Their analysis highlights that strain transfer improves with increasing wavelength and is nearly ideal for typical seismic wavelengths. The first step, which we refer to as cable-ground coupling, depends on the contact points between cable and ground, the cable geometry in between contacts, both depending on the properties of the cable and the ground surface. In buried configurations with tight ground contact and no slip, coupling is assumed to be ideal—the cable moves with the ground. The no-slip assumption is discussed below. In that case, the strain transfer from the ground to the buried cable is purely determined

by the cable-to-fiber transfer. In unburied setups, however, the cable only follows ground motion at contact points, but not necessarily on the segments in between. Understanding coupling mechanisms is crucial for generalizing results, predicting performance under lunar conditions (e.g., reduced gravity), and designing optimized cables. We hypothesize that the main coupling mechanism is bending stress relief described in the following. In our unburied deployment in the LRS the cables are draped on the surface as flat as possible, still there are usually a few mm to cm between individual contact points with the ground. In between those contact points the cables are suspended in the air due to their stiffness and slightly arching towards the ground due to gravity. When the ground deforms and neighboring contact points move toward each other, the cable segment suspended between them can bend perpendicularly to the strain direction (as illustrated in Figure 6). This bending allows the cable to relieve part of the stress rather than transferring it as axial strain to the fiber core. While a full quantification of this effect is beyond the scope of this study, beam theory suggests that the resistance to bending increases with both the Young’s modulus and the fourth power of the cable diameter. This implies that thinner and more flexible cables bend more easily under horizontal shortening, thereby relieving more stress through bending and reducing the strain transferred to the fiber. This aligns with our experimental observations, where thinner cables consistently recorded lower amplitudes, indicating less efficient strain transfer. In addition to cable stiffness and diameter, the length of the suspended segment also plays a critical role: longer segments bend more easily under the same horizontal deformation, further reducing strain transfer. This aligns with our observation of decreased amplitudes for all cable types with increasing undulation length. When free-hanging segments become too long, resonance effects may dominate, further degrading strain transfer and introducing phase shifts.

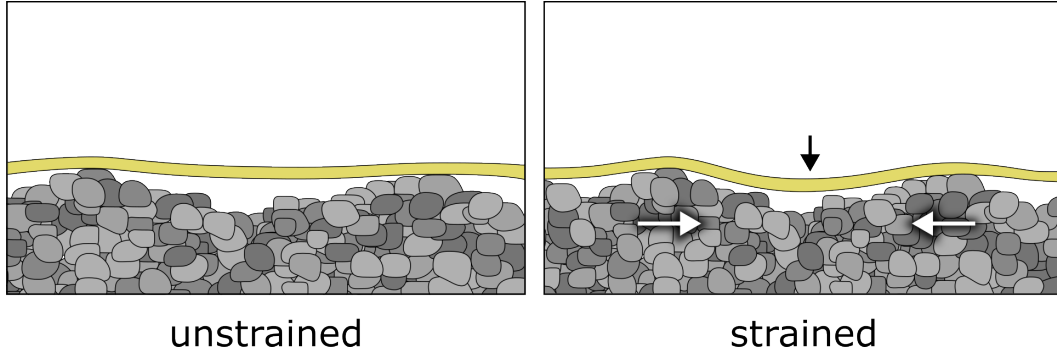


Figure 6. Conceptual illustration of the bending stress relief mechanism proposed to explain reduced strain transfer in unburied DAS cables. When ground deformation causes contact points to move closer or further apart, suspended cable segments bend rather than compress or stretch, relieving stress and reducing axial strain transmission to the fiber core.

Slip between the cable and the ground is another potential mechanism affecting coupling. Slip occurs when the force required to deform the cable exceeds the frictional force at the contact points. If slip were to occur, it would be amplitude-dependent: small deformations could be sustained by friction, while larger deformations might exceed the frictional threshold, leading to slip. This would imply a non-linear coupling behavior at the onset of slip. To assess whether slip is likely under realistic seismic conditions, we compared the frictional force at the cable-ground contact to the deformation force required to deform the cable. The frictional force is estimated as $F_f = \mu \cdot m_c \cdot g \approx 0.07 \text{ N/m}$, where $\mu \approx 0.5$ is the friction coefficient which can vary depending on the surface properties, $m_c = 0.015 \text{ kg/m}$ is an average cable mass per unit length, and g is gravitational acceleration. The deformation force, calculated using Hooke’s law and assuming strain

amplitudes typical of seismic waves, is given by $F_d = \frac{2AE\varepsilon_{\max}}{\lambda} \approx 0.0012 - 0.012$ N for small strains. For stronger, active sources, this could increase to the range of 0.05–0.5 N. To summarize the slip condition, we define a dimensionless number comparing the available frictional force to the required deformation force:

$$\Gamma = \frac{F_f}{F_d} = \frac{\mu \cdot m_c \cdot g \cdot \lambda}{2AE\varepsilon_{\max}}, \quad (1)$$

where $\Gamma > 1$ indicates that friction is sufficient to prevent slip. For the parameter ranges considered, $\Gamma > 1$, suggesting that slip is unlikely under most realistic seismic scenarios. This conclusion is consistent with our experimental observations: we did not detect any non-linear behavior indicative of slip, even at amplitudes larger than those expected from natural seismic events. However, slip may still be relevant under conditions with low friction, such as due to substrate or cable properties, or under reduced lunar gravity, which lowers the normal force and thereby the available friction. Harmon et al. (2024) hypothesize that slip may have occurred in their unburied DAS experiment due to the low-friction properties of the cinder and ash substrates, and suggest that such slip can act like attenuation and cause frequency-dependent phase shifts.

4.3 Implications for lunar DAS

Our results indicate that unburied DAS may be viable for lunar applications, especially with robust cables like the OD5.5. Both OD5.5 and OD3 maintained relatively constant amplitude and phase response when unburied, suggesting reliable strain transfer in the tested frequency range. However, the detrimental effects of free-hanging segments underscore the importance of careful deployment. We recommend unspooling the cable onto the surface and pressing it down, e.g., by rolling a wheel over it, to ensure continuous contact. This simple method could significantly enhance coupling, as even minor deployment variations had substantial effects in our experiments. A similar concept has been proposed by Burns et al. (2023), who suggest rolling out a flat tape tether for lunar deployment. In our experiments, the thicker and stiffer OD5.5 consistently showed better coupling and higher recorded amplitudes. This observation is consistent with the predictions of our hypothesized coupling mechanisms. Our results also show that wind noise is a major factor degrading signal quality in unburied DAS on Earth, particularly for lighter cables like the OD0.9. This issue would not be present on the Moon due to the absence of an atmosphere, representing a clear advantage for lunar deployments. An important consideration for lunar deployment is the effect of reduced gravity on cable-ground coupling. We observed that heavier and stiffer cables, such as the OD5.5, exhibited better coupling quality. If cable weight is indeed a dominant factor in achieving good coupling, then replicating this behavior on the Moon, where gravity is approximately one-sixth that of Earth, might require cables that are six times heavier to exert the same normal force on the ground. However, it remains unclear whether weight alone governs coupling quality. Other factors such as cable diameter and stiffness may play equally or more significant roles, particularly if bending stress relief is indeed a relevant factor. Since these properties are independent of gravity, they would remain unchanged on the Moon. This uncertainty highlights the need for a deeper understanding of the physical mechanisms behind strain transfer.

Open Research Section

The used data sets and data processing codes are openly available at the ETH Zurich Research Collection via <https://doi.org/10.3929/ethz-c-000785214> (Probst et al., 2025).

Conflict of Interest disclosure

The authors declare there are no conflicts of interest for this manuscript.

Acknowledgments

SP and JOAR acknowledge funding from the Swiss National Science Foundation, grant 200021_212064/1. CD was supported by the Los Alamos National Laboratory (LANL) through its Center for Space and Earth Science (CSES). CSES is funded by LANL's Laboratory Directed Research and Development (LDRD) program under project number 20240477CR. AZ acknowledges support from: LANL Directed Research and Center for Space and Earth Sciences 20240065DR/YW0300. We thank LASN, Peter Roberts, and Emily Rodriguez for the use of seismometers and for helpful insight, and William Redford for measurements of wind speed. This manuscript has been assigned the number LA-UR-25-29704 by Los Alamos National Laboratory.

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Controlled Source DAS Coupling Tests: Implications for Unburied Deployment on the Moon and Earth

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Key Points:

- Unburied distributed acoustic sensing can be feasible on the Moon with suitable cable choice and careful deployment
- Thick unburied cables exhibit flat amplitude and phase spectra, performing comparably to buried cables
- Wind and surface undulations degrade signal quality in unburied terrestrial deployments, especially for thin cables

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Abstract

Moonquake recordings are key to understanding the Moon’s internal structure, yet existing data from the Apollo missions are limited. Distributed Acoustic Sensing (DAS) offers a promising alternative to conventional seismometers for future lunar seismic monitoring, particularly due to its robustness and suitability for high-scattering environments like the lunar regolith. Since burial of DAS cables may be impractical on the Moon, understanding the coupling performance of unburied cables is critical. In this study, we present experimental results from controlled shaker tests in a lunar regolith simulant, comparing the coupling performance of buried and unburied fiber-optic cables with varying mechanical properties and assessing the impact of wind noise and surface undulations. Our findings show that burial generally improves signal amplitude and phase reliability, while unburied cables yield weaker signals and, in some cases, exhibit phase shifts. However, thicker and stiffer unburied cables achieve comparable performance as buried cables, maintaining stable amplitude ratios and phases. Furthermore, we demonstrate that free-hanging cable segments degrade coupling quality substantially for all tested cables, emphasizing the importance of continuous ground contact. We also show that wind is a major contributor to poor signal quality in unburied DAS deployments on Earth - an issue that would not arise on the Moon due to its lack of atmosphere. These results suggest that with appropriate cable selection and deployment strategies, unburied DAS is a viable option for lunar seismic applications.

Plain Language Summary

Understanding the Moon’s interior is important for planetary science, and seismic measurements are one of the most effective tools for this purpose. These measurements rely on detecting moonquakes: vibrations that travel through the lunar subsurface. However, the only available moonquake data were collected during the Apollo missions using a small number of traditional seismometers. Distributed Acoustic Sensing (DAS) is a newer technique that could improve future lunar seismic monitoring. DAS uses a fiber-optic cable to measure ground motion along its entire length, effectively turning the cable into thousands of vibration sensors. This method offers dense spatial coverage and could simplify deployment compared to conventional instruments. On Earth, DAS cables are typically buried to ensure good contact with the ground. However, on the Moon, burial may not be feasible. To investigate whether unburied cables can still provide reliable data, we conducted controlled experiments using a lunar soil simulant and several types of fiber-optic cables. We found that thicker, stiffer cables can record stable signals even when unburied, provided they maintain contact with the ground. These results suggest that DAS could be a viable method for lunar seismic sensing, enabling large-scale measurements without the need for complex installation procedures.

1 Introduction

Currently, the only seismic dataset recorded on the Moon originates from the Apollo missions, during which both active and passive seismic experiments were conducted. Five stations deployed on the near side of the Moon recorded over 12,000 seismic events from 1969-1977 (see Nunn et al. (2020) for a review of the instrumentation and collected data). This dataset has led to the identification of four distinct types of natural seismic sources on the Moon, namely, deep moonquakes, shallow moonquakes, thermal moonquakes, and meteoroid impacts (Nakamura et al., 1982). The collected seismic data have been the primary source for estimating the lunar seismic velocity structure and have been extensively analyzed over the decades and continue to receive fresh scrutiny, for example, by Sollberger et al. (2016), who used spatial wavefield gradients to construct an elastic model of the shallow lunar subsurface. A major challenge in interpreting moonquake seismograms is the presence of long-duration and high-amplitude coda waves that are much more

pronounced than exhibited in earthquake seismograms. Caused by the strongly scattering and low attenuating lunar shallow subsurface, the coda waves obscure later arriving seismic phases.

An approach that could help address this challenge is an emerging sensing technology called Distributed Acoustic Sensing (DAS), which measures dynamic strain over long distances. DAS has rapidly gained traction in a broad range of geophysical applications as an alternative to conventional ground motion sensors such as geophones and seismometers. A DAS setup consists of a fiber-optic cable connected to an interrogator. The interrogator sends laser pulses into the optical fiber and measures the returning light that experiences Rayleigh back-scattering from naturally occurring inhomogeneities in the fiber smaller than the probing wavelength. This backscattered light undergoes phase changes as the scatterers move due to strain on the fiber, consequently, the dynamic strain of the optical fiber can be inferred, thereby transforming the optical fiber cable into a dense array of vibration sensors. As a result, DAS is sensitive to a length change of the cable which - in the case of perfect coupling of cable to the ground - corresponds to the inline strain.

DAS is a technology that is robust and easy to deploy, making it a promising candidate for lunar deployment. A single fiber-optic cable attached to an interrogator can record over tens of kilometers and replace thousands of sensors, eliminating the need for individual power supplies or data acquisition systems. Fiber-optic sensors can withstand harsh environmental conditions, including extreme temperatures and radiation. They have been successfully deployed in environments ranging from cryogenic temperatures (Yang et al., 2021) to temperatures as high as 1000°C in ϕ -OTDR-based DAS experiments (Xu et al., 2025). J. Wu et al. (2021) demonstrated that OFDR-based fiber sensors maintained stable Rayleigh backscattering profiles over multiple months inside a nuclear reactor core, exposed to both high temperatures and neutron radiation. Additionally, long-duration space exposure experiments have shown that fiber-optic cables can remain functional after years in orbit, even when unshielded (e.g., Hartmayer et al., 1993). This study confirms that cables can endure temperature cycling, radiation exposure, and polymer aging in space. However, planned mission duration is critical since radiation effects and material degradation accumulate over time, and the small but non-negligible risk of micrometeoroid impacts on the cable and a resulting fiber breakage increases with exposure time. Recent modeling studies have demonstrated the potential of DAS for lunar applications (W. Wu et al., 2024). The dense channel spacing that DAS offers is particularly suited for high-scattering environments such as the lunar shallow subsurface and could allow DAS to recover deep seismic phases that were obscured in Apollo data due to scattering. Zhai et al. (2024) estimate that DAS could detect around 60 % of the moonquakes observed by the Apollo seismometers. However, with foreseeable improvements in DAS technology this number could rise to detect nearly all moonquakes previously observed by Apollo. Their findings emphasize the critical role of DAS performance in moonquake detectability, underscoring the relevance of understanding strain transfer and amplitude behavior as investigated in this study.

However, several open questions need to be addressed before DAS can be deployed on the Moon. A key challenge is determining how to best deploy the fiber-optic cable ensuring good coupling. On Earth, burial is commonly used to ensure good coupling, and studies have shown that it significantly improves signal quality compared to unburied deployment (Dou et al., 2017; Rossi et al., 2022). However, burial may be infeasible in the lunar environment. Unburied deployments on Earth often exhibit poor signal quality, which has been partly attributed to atmospheric acoustic noise, particularly wind and thunder (Carr et al., 2025; Viens et al., 2025; Hudson et al., 2021). This atmospheric noise is not expected to be a significant source of noise, as the Moon has only a thin exosphere. However, it is still unclear whether the poor performance of unburied fiber is primarily due to poor ground coupling or rather due to atmospheric noise. Understanding this distinction is critical for assessing whether unburied deployment on the Moon can yield satisfactory seismic data.

This study employs lunar regolith simulant (LRS) to examine the relationship between the conditions of a fiber optic cable deployment and data quality, highlighting implications for lunar exploration. We are building upon the work of Zandanel et al. (n.d.) which used the same experimental setup to record local and regional earthquakes and found that unburied fiber is capable of detecting earthquakes with a weaker amplitude than buried fiber, but with only a modest decrease in the signal-to-noise. This study aims to quantify this effect and investigate whether the observed amplitude attenuation factor is frequency-dependent and if the phase recorded with unburied fiber is reliable. Using a shaker source to produce controlled and repeatable ground motion in the LRS, we examined the coupling of different cable types with varying mechanical properties, such as outer diameter and weight. Employing cables with drastically different properties is motivated by two key considerations: (1) comparing the performance of cables with differing characteristics can help elucidate the physical mechanisms that govern fiber-regolith coupling and how these mechanisms depend on cable properties, and (2) for lunar deployment, weight and robustness are critical factors. Due to the high cost of transport to the Moon, lightweight and robust cables are highly desirable, especially given that deployment strategies are still under development, with potential concepts of launching the cable onto the surface. Additionally, we investigated the impact of ambient wind to understand how much wind noise contributes to the poor signal quality of unburied fiber observed on Earth. Lastly, we investigated the impact of small-scale surface topography on coupling quality to assess the robustness of unburied cables to surface heterogeneities.

2 Methods

2.1 Experimental setup

An experiment was built that mimics some of the conditions that are present on the Moon. A 9.5-meter-long trough was filled with a generic lunar regolith simulant (LRS), composed of crushed basalt (Figure 1c). The trough was filled to a depth of approximately 59 cm with the LRS, forming a flat surface. A detailed characterization of the LRS is provided in Zandanel et al. (n.d.).

Systematically investigating the coupling behavior of various cable types and deployment configurations could only be achieved by exciting the LRS with controlled frequencies and amplitudes. To generate such controlled ground motion, a shaker (Brüel & Kjær 4809) was mounted at the short end of the trough, aligned with the LRS surface, and oriented to excite motion along the longitudinal axis (see Figure 1d). The shaker was driven with a sequence of single-frequency signals ranging from 3 to 30 Hz in 1 Hz increments. Each frequency was excited for 250 periods and repeated at 10 different amplitude levels. Frequencies below 3 Hz were excluded due to shaker limitations, and frequencies above 30 Hz were excluded to avoid the resonance of the trough, which had the first mode at approximately 30 Hz (see Figure 1b). The energy of most natural moonquakes is below 20 Hz (Civilini et al., 2023).

In addition, one experiment simulated an earthquake by driving the shaker with a real earthquake recording as input. We collected DAS data with three cables that exhibit strongly differing mechanical properties: a tight buffered cable with an outer diameter (OD) of 5.5 mm (referred to as OD5.5), a tight buffered cable with an OD of 3 mm (OD3), and a jacketed cable with an OD of 0.9 mm (OD0.9). The properties of those cables are summarized in Table 1.

To investigate the impact of burial on coupling, we deployed the cables at three depths in the LRS: unburied (laid directly on the surface without additional coupling), shallow-buried (at a depth of approximately 2 to 3 cm below the LRS surface), deep-buried (at a depth of approximately 11 cm) (see Figure 1a). Each cable type was deployed in both unburied and shallow-buried configurations. The OD0.9 was deployed in two separate lengths for each of these configurations, providing two independent measurements per depth. The OD5.5, which contains two optical fibers within a single cable, was interro-

gated on both fibers, yielding two measurements from the same physical cable. Only the OD3 was additionally deployed in the deep-buried configuration. To test the robustness of the results and assess the sensitivity of coupling to small variations in deployment, all unburied cables were redeployed after one round of data acquisition. This involved removing and repositioning the fibers on the surface, followed by a repeated acquisition under the same conditions.

DAS data were acquired using a Terra15 Treble+ interrogator at a sampling rate of 3 kHz, with a gauge length and pulse length of 2.45 meters and a channel spacing of 0.82 meters. This configuration was chosen to balance spatial resolution and signal-to-noise ratio. For comparative analysis of the cables and deployment types, we used the central DAS channels at the midpoint of the trough.

A seismometer (Trillium compact 120s) served as a ground-truth reference and was placed at the center of the trough, aligned with the DAS channels and sampled at 250 Hz.

The trough was located indoors, shielding it from natural wind. To simulate wind noise, a fan was positioned approximately 30 cm above the cables at the center of the trough, blowing vertically downward. The fan operated at three intensity settings, corresponding to maximum wind speeds of 2, 2.4, and 3 m/s at a distance of 30 cm away from the fan. Those wind speeds, classified as a "light breeze" on the Beaufort wind scale, are typical of outdoor conditions on Earth.

To assess the impact of surface irregularities, we introduced artificial undulations in the trough by forming grain piles of varying spacings (15-25 cm, 30-50 cm, 60-100 cm). Cables were laid across these piles such that they contacted the ground only at the peaks and were suspended in between (see Figure 1e). The shaker experiment was repeated for each undulation configuration.

Table 1. Specifications of the used fiber-optic cables. OD = outer diameter. SM = single-mode fiber. MM = multi-mode fiber. Cable OD5.5 contains both single-mode and multi-mode fibers, but only the single-mode fibers were interrogated for the measurements.

Cable type	Abbrev.	OD (mm)	Nr. of cores	Buffering	Weight
PS-2S2M-1PU055-01	OD5.5	5.5	2 SM, 2 MM	tight buffered	27 g/m
SMF-28-100	OD0.9	0.9	1 SM	yellow Hytrel tubing	3.1 g/m
Silixa	OD3	3	1 SM	tight buffered	8.6 g/m

2.2 Data processing

To compare the fiber segments, we analyzed the DAS signal recorded at the central position of each segment. A tap test was performed at the midpoint of the trough to identify the central positions, which were localized with an estimated accuracy of half the channel distance. Subsequently, the signal at the central position was interpolated by applying a weighted average of the two adjacent channels. The amplitude response of each fiber segment was characterized as a function of both excitation frequency and amplitude of the shaker. For each excitation frequency, the corresponding time window was divided into 1-second segments. The frequency spectrum of each segment was computed and averaged to obtain a representative spectrum. This approach ensured consistent spectral resolution and comparability across all excitation frequencies. From each averaged spectrum, the amplitude and phase at the excited frequency were extracted. The seismometer data were processed using the same spectral analysis workflow as the DAS data.

This procedure allowed us to investigate how the recorded amplitude, and thus the coupling efficiency between the fiber and the ground, varies with cable type, deployment depth,

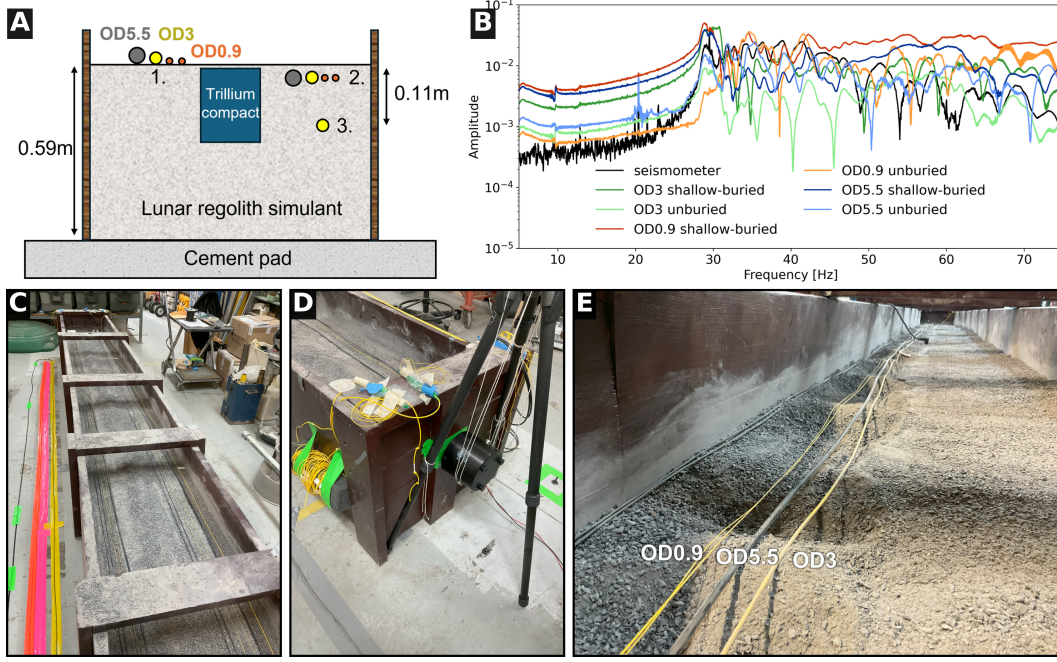


Figure 1. Overview of the experimental setup and conditions. (A) Schematic cross-section of the trough showing the three deployment depths of the fiber-optic cables 1) unburied, 2) shallow-buried, and 3) deep-buried. (B) Amplitude spectrum of a shaker sweep recorded on different fiber segments with minimal undulations, with excitation ranging from 5 to 150 Hz. Resonance behavior present at frequencies above 30 Hz. For comparison, the amplitude spectrum of the co-located seismometer is also shown (normalized for visual clarity), illustrating that the resonance is a property of the experimental trough rather than the fiber coupling. (C) Photograph of the 10-meter-long trough filled with lunar regolith simulant (LRS) and deployed cables. (D) Shaker mounted at the short end of the trough for controlled excitation. (E) Surface undulations created by grain piles (spaced 15-25 cm apart), with cables suspended between contact points to simulate irregular topography.

strain amplitude, and frequency. The amplitudes were divided by the amplitudes measured by the seismometer to account for frequency-dependent variations in shaker efficiency, yielding amplitude ratios. To summarize the coupling performance of each cable and deployment configuration with a single metric, we computed an average amplitude factor. This factor was obtained by averaging the amplitude ratios across all excitation frequencies and amplitudes. Additionally, for the OD5.5 and OD0.9 where two fibers of each cable could be interrogated at the same point - either from two fibers within the same cable (OD5.5) or from two separate cable lengths (OD0.9) - the results from both measurements were averaged to obtain one representative value. An exception is OD0.9 unburied for which the results from both fibers were processed separately, since their performance was very variable, as discussed in more detail later on. The uncertainty of the amplitude response was estimated by accounting for the localization uncertainty of the central channel. Specifically, the analysis was repeated after shifting the central position by ± 0.5 channel spacing, and the resulting variation in amplitude was used to quantify the uncertainty. This channel location uncertainty was considered the dominant source of error, as the shaker was mounted at one end of the trough, resulting in

a strong amplitude gradient along the trough axis. Consequently, even small deviations in channel position could lead to significant differences in recorded amplitude. For the analysis of wind-induced noise, we also focused on the DAS signals recorded at the midpoint of the trough where the fan was positioned. The wind noise data were processed by calculating the power spectral density (PSD) of each trace using Welch's method, with a window length of approximately 3 seconds. This allows us to assess the wind's impact across different frequency bands. To quantify the average intensity of wind-induced vibrations in the seismic frequency band, each trace was further filtered with a fourth-order Butterworth low-pass filter with a cutoff frequency of 20 Hz. The mean absolute amplitude of the filtered signal was then computed in the time domain, providing a representative measure of the sub-20 Hz strain rate noise for each cable segment and wind condition. Here again, unburied cables were redeployed after one round of data acquisition.

3 Results

3.1 Comparison of cable types and deployment

Figure 2a and 2b illustrate how each cable and deployment type responded to shaker excitation, in terms of amplitude and phase. To assess DAS coupling performance, we compare the recorded amplitudes to those of a colocated seismometer, which serves as a ground-truth reference. The seismometer is assumed to have a flat amplitude and phase response across the tested frequency range. Literature suggests that DAS also exhibits flat amplitude and phase spectra when well coupled (Paitz et al., 2021), and this assumption can be verified through comparison with the seismometer. It is important to note that amplitude ratios are not expected to be unity even under ideal coupling conditions, because DAS measures strain rate while the seismometer measures ground velocity. Theoretically, conversion approaches exist to convert DAS recordings to velocity (e.g., Trabattini et al., 2023). However, such conversions assume ideal coupling and long linear cable segments - conditions not met in our experiment. Therefore, we refrain from converting DAS and the seismometer to the same units and instead perform a qualitative comparison using amplitude ratios. The absolute maximum ground particle velocities recorded by the seismometer were on the order of $1\text{e-}4$ m/s. Illustration of the maximum velocities across frequencies and shaker amplitudes is included in the supplementary material.

Shallow and deep-buried fiber segments exhibit relatively constant, frequency-independent amplitude responses. Their phase responses are flat and consistent with the seismometer, indicating reliable strain measurements in the tested frequency-range aside from a uniform amplitude scaling. Unburied fiber segments OD5.5 and OD3 also exhibit stable amplitude factors and flat phases with the exception of an observed phase shift at 25 Hz for OD5.5. Comparison of the two deployed unburied segments of OD0.9 and between repeated deployments OD0.9 shows relatively flat phase and amplitude behavior in some cases, while under less favorable conditions, it exhibits erratic amplitude and phase behavior. In Figure 2, both OD0.9 unburied fiber segments are shown separately to highlight the variability observed between deployments. Repeatability and deployment sensitivity are further discussed in Section 4.1. For one of the unburied OD0.9 segments, a pronounced phase shift is observed at frequencies above 13 Hz, which may indicate that the fiber is resonating at those frequencies.

A comparison across different shaker-driving amplitudes shows a linear relationship between input and recorded amplitude, with no evidence of nonlinear behavior (see supplemental material). To summarize the coupling performance of each cable and deployment configuration with a single metric, we computed an average amplitude response, shown in Figure 2c. For buried cables, the values are relatively similar, with OD0.9 showing the highest amplitude, but still close to the uncertainty range of the other cable types.

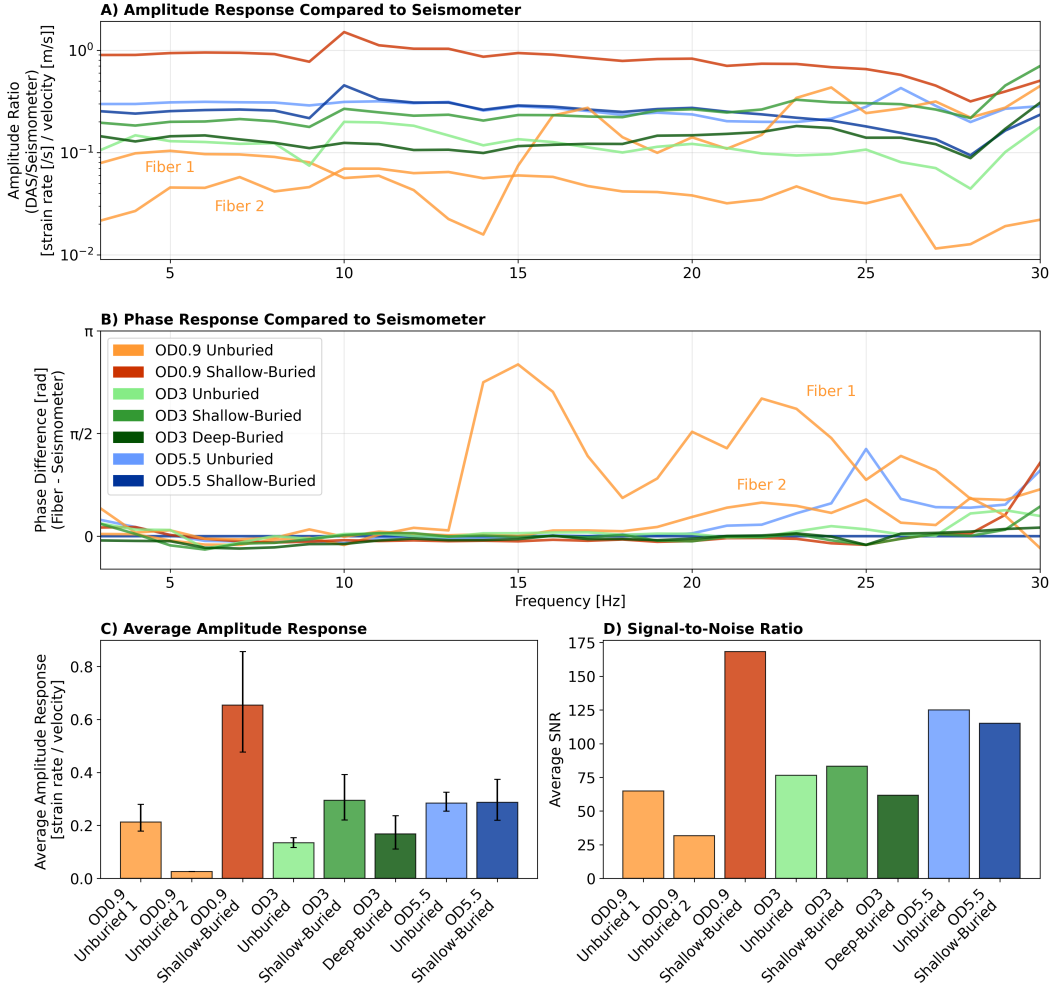


Figure 2. Fiber-optic cable coupling efficiency for different cable types and deployment configurations. (A) Amplitude response showing the ratio of DAS amplitude compared to seismometer amplitude across frequencies 3-30 Hz for shaker amplitude level 7. (B) Phase response showing phase differences of fiber segments relative to the seismometer. (C) Coupling efficiency summary showing amplitude factors averaged across all excited frequencies and amplitudes, with error bars indicating variability due to channel location uncertainty. (D) Signal-to-noise ratio of fiber segments averaged over all excited frequencies and amplitudes. For the OD0.9 unburied configuration, both deployed fibers are shown in all subfigures to illustrate the observed variability between deployments. In contrast, other cables with multiple recordings showed close agreement and are represented by a single curve or value.

In contrast, unburied cables generally exhibit lower amplitude (roughly half that of buried cables), except for OD5.5, which shows comparable values in both configurations. In addition to amplitude and phase analysis, we computed the signal-to-noise ratio (SNR) for each fiber segment, shown in Figure 2d. The SNR was calculated by comparing the signal amplitude during excitation to the noise amplitudes. The SNR trends largely mirror the average amplitude responses, but with some notable differences. OD5.5 consistently shows higher SNR than OD3 in both buried and unburied configurations. Unburied

OD3 achieves SNR values comparable to its buried counterpart, while OD0.9 exhibits significantly lower SNR when unburied.

3.2 Impact of surface undulations

In field deployments, ensuring that fiber-optic cables lie flat and maintain continuous ground contact is labor-intensive. Faster, less controlled deployments may result in cables spanning small surface undulations, potentially degrading coupling quality. To investigate this effect, we introduced artificial surface undulations of varying sizes in the trough. In these deployments, the cables are only in contact with the LRS at the peaks of the undulations. Figure 3 illustrates the impact of these undulations on the signal quality. Coupling quality decreases significantly when cables are suspended across undulations rather than in continuous contact with the ground. For small undulations, the OD0.9 already shows a marked reduction in signal quality across all tested frequencies. In contrast, the OD5.5 and the OD3 maintain relatively good coupling at frequencies below 10 Hz but also exhibit poor performance at higher frequencies. For larger undulations, all unburied cables show substantial signal degradation, characterized by low amplitudes and strong variability. Additionally, phase responses become erratic, with large and inconsistent phase shifts. These drastic phase shifts, especially at higher frequencies may indicate fiber resonance. Notably, the onset frequency of these phase shifts decreases as the undulation spacing increases, suggesting that longer free-hanging segments lower the resonance frequency.

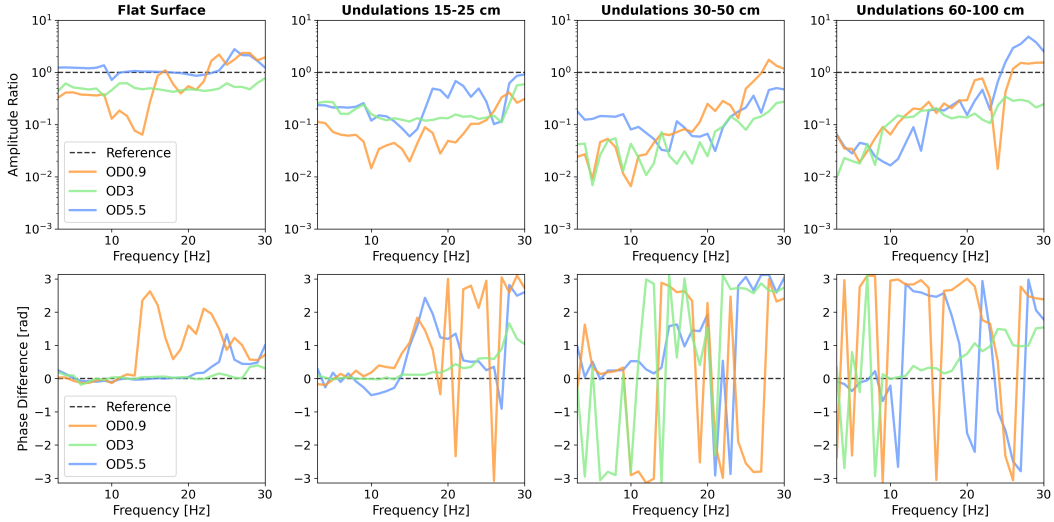


Figure 3. Impact of surface undulations on the coupling of unburied cables (shaker amplitude level 7). Top row: amplitude response plots; bottom row: phase response plots. The leftmost column represents the flat (reference) case (as in Figure 2a and 2b), while the subsequent columns show increasing undulation spacing. All plots compare unburied cables to the same reference: shallow-buried OD5.5. As undulation spacing increases, coupling quality progressively degrades, evidenced by reduced amplitudes and phase shifts, particularly in the two cases with the largest undulations.

3.3 Impact of wind noise

To assess the influence of wind on DAS signal quality, we compare the noise floor of fiber segments exposed to artificially generated wind with those without it. Since the

trough is located indoors, the fibers are naturally protected from ambient wind, allowing for controlled testing. Wind noise might contribute significantly to the poor signal quality observed in unburied cables deployed outdoors. To quantify the contribution of wind on the poor signal quality observed with unburied fiber, we used a fan to simulate wind and analyze the resulting increase in noise levels. The results are presented in Figure 4 (for improved visibility, the spectra are only shown up to 200 Hz). The shallow-buried fibers are practically unaffected by wind, confirming their robustness against wind noise. In contrast, all unburied fibers exhibit elevated noise levels, though the degree of impact varies significantly between cable types. OD5.5 is the least affected, with wind-induced noise primarily below 200 Hz and maximum amplitudes of the wind noise reaching approximately -130 dB/Hz. OD3 shows increased sensitivity, with noise extending up to 400 Hz and peak amplitudes around -125 dB/Hz. OD0.9 is the most affected, with wind noise up to 700 Hz. Particularly strong wind amplitudes are observed in bands between 20-30 Hz and 400-600 Hz, with maximum amplitudes reaching -110 dB/Hz. For seismological applications, the frequency band below 20 Hz is of particular relevance. In this band, the wind-induced noise amplitudes at fan speed 3 are $1.99\text{e-}07$ m/m/s for OD5.5, $7.15\text{e-}07$ m/m/s for OD3, and $2.57\text{e-}06$ m/m/s for OD0.9, sufficient to render moderate earthquakes undetectable.

To illustrate the impact of wind noise on seismic signal detectability, we recorded a shaker-generated simulated earthquake with and without wind noise from the fan. As shown in Figure 5, the presence of wind noise significantly affects the signal detectability. On OD0.9 and OD3 unburied, the event becomes undetectable while on OD5.5 unburied, the signal-to-noise ratio is noticeably reduced. The PSDs of the recordings show similar noise patterns as Figure 4 and are included in the supplementary material.

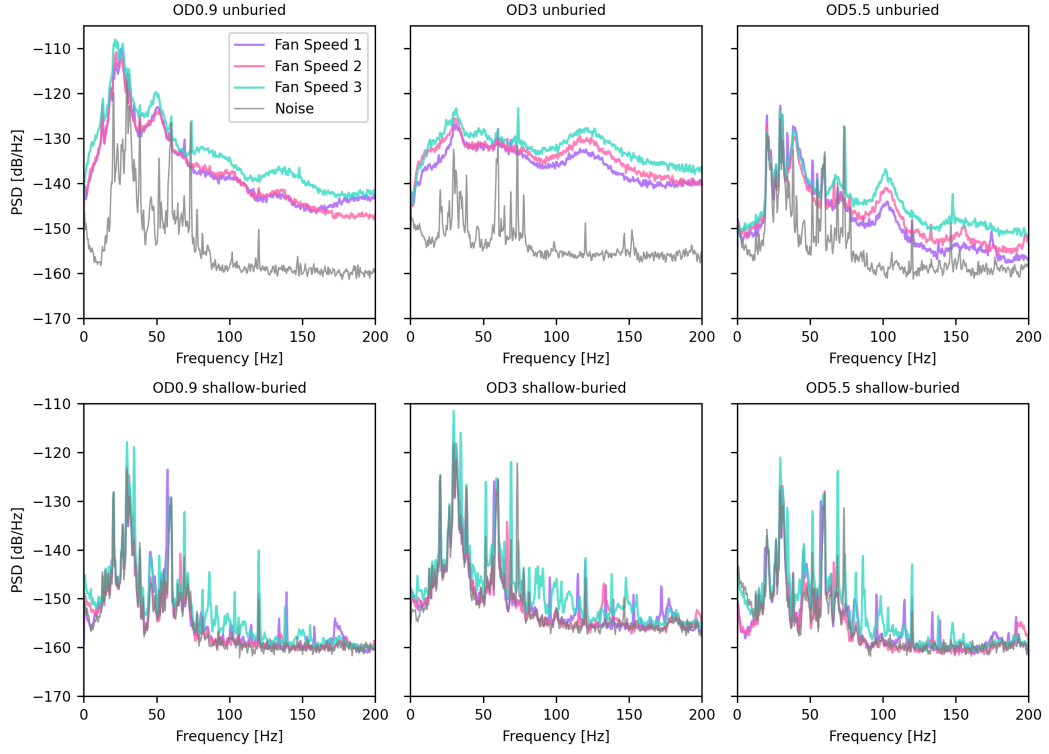


Figure 4. Power spectral density (PSD) comparison of wind-induced noise across different cable segments. Top row: unburied cables (OD0.9, OD3, OD5.5); bottom row: corresponding shallow-buried segments. Each plot shows PSD curves from 0-200 Hz for the four conditions. Fan speeds 1 to 3 correspond to the fan blowing directly onto the cables at increasing wind settings, while the "Noise" condition was recorded with the fan operating but directed away from the cables. This setup captures the ambient noise and fan motor noise without wind impact on the fibers. The difference between the "Noise" and fan speed curves isolates the effect of wind on signal quality.

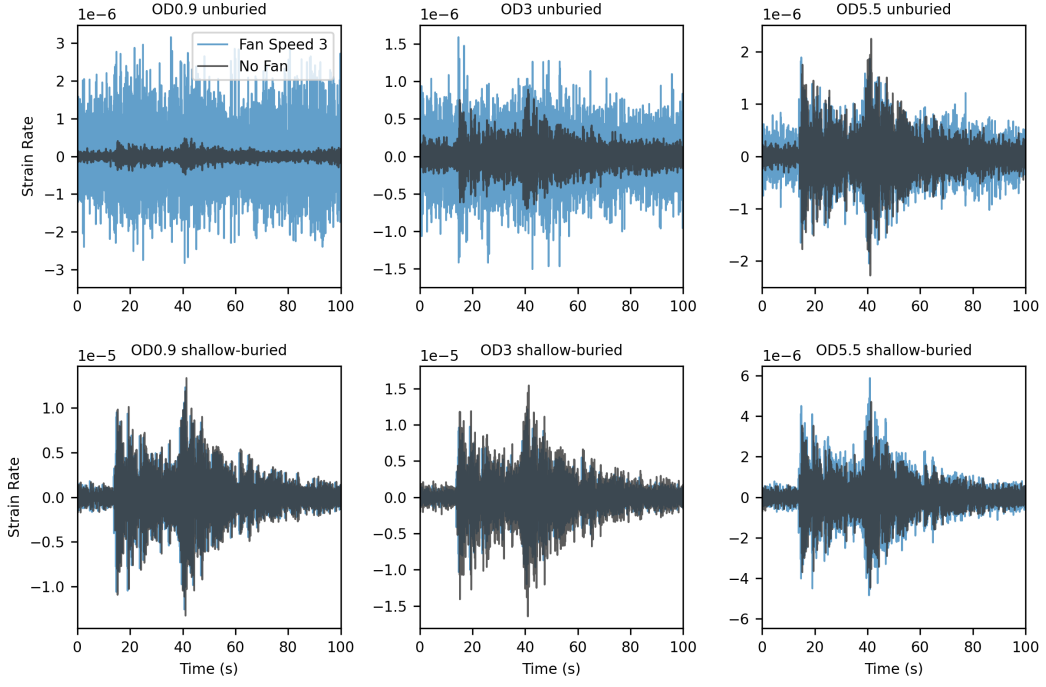


Figure 5. Comparison of simulated earthquake signals recorded on different fiber segments with and without wind noise. The signals were generated with the shaker to simulate a seismic event. Black lines show recordings in the absence of wind, while blue lines show recordings with the fan operating at fan speed 3. All traces were bandpass filtered between 3 and 15 Hz. The simulated earthquake is clearly visible in all traces without wind, although with lower signal strength on OD0.9 unburied and OD3 unburied. With wind present, the seismic signal becomes completely obscured in the OD0.9 and OD3 unburied traces, while it remains visible but with reduced SNR on the OD5.5 unburied trace. All shallow-buried fiber segments show minimal effect from wind noise.

4 Discussion

4.1 Discussion of results

Buried cables consistently recorded signals with a stable amplitude factor and correct phase. Differences of the amplitude factor between cable types were within a factor of two. The deep-buried cable showed slightly lower amplitudes than the shallow-buried ones, but the difference is within the uncertainty range. Since only one cable was tested in the deep-buried configuration, no definitive conclusions can be drawn about the effect of burial depth.

Cable type had a more pronounced impact in unburied deployments. Among the three tested cables, the two thicker ones (OD5.5 and OD3) recorded amplitudes comparable to their buried counterparts and maintained reliable phase information across most frequencies. In contrast, the thinner OD0.9 cable exhibited significantly lower amplitudes and high variability depending on deployment details. Some OD0.9 deployments still yielded correct phase data, albeit with low amplitudes, while others produced highly variable amplitudes and phase disparity. After redeploying the unburied cables for the repeated test, the OD0.9's signal quality degraded noticeably. In the initial deployment, the cables had been in place for some time before data acquisition, allowing them to settle slightly into the fine surface dust and conform more closely to the ground, which likely improved coupling. After redeployment, the cables did not follow the ground as closely, often being slightly lifted by their own stiffness, resulting in reduced contact and poorer coupling. These observations suggest that thin cables are particularly sensitive to small variations in deployment geometry and surface conformity.

The SNR analysis supports these findings. For OD5.5 and OD3, SNR values were similar between buried and unburied deployments, indicating that signal detectability remains robust even without burial. In contrast, OD0.9 showed a substantial reduction in SNR when unburied, consistent with its lower amplitude and higher variability. These results align with the findings of Zandanel et al. (n.d.), who observed reduced amplitudes but comparable SNR for unburied OD3 compared to buried OD3 in earthquake recordings. It is important to note that the noise in our experiment is predominantly environmental and not purely instrumental: Since the noise itself is influenced by the same coupling mechanisms as the signal, the SNR trends partly reflect signal-to-signal comparisons rather than signal-to-instrumental-noise ratios. In a quieter environment like the lunar surface, where environmental noise is absent, the relative SNR differences between cable types may be more pronounced.

Continuous ground contact is critical for signal quality in unburied deployments. Surface undulations that cause cables to hang freely degrade signal quality, leading to unreliable amplitudes and phases. Even short free-hanging segments (~ 20 cm) negatively affected all cable types, with the impact increasing with undulation length. A plausible explanation is that these segments resonate when excited, with resonance frequencies depending on cable properties and undulation length. This interpretation is supported by the observed drastic phase shifts in the presence of undulations. Additionally, some resonance-like behavior was already observed for the unburied OD0.9 even in the absence of undulations, suggesting that this thinner cable is particularly prone to resonating when not well coupled. Resonance effects in DAS cables have also been reported in previous studies, such as Hudson et al. (2021), who observed wind-induced cable resonance in unburied deployments.

Deployment technique significantly influences signal quality. Pressing cables down to ensure ground contact reduces free-hanging segments and improves coupling. This aligns with findings by Mjehovich et al. (2023), who observed improved signal quality when cables were pressed into snow compared to deployment without further coupling efforts, and by Harmon et al. (2022), who reported poor coupling on grassy surfaces where cables were suspended above the ground by the grass.

We found that wind noise significantly increases the noise floor for unburied cables. The OD0.9 was most affected, followed by the OD3 and then the OD5.5. This trend corre-

lates with cable diameter and mass: the lighter OD0.9 was more susceptible to wind-induced motion, which was visibly observed during testing, unlike the heavier OD5.5. However, all cable types exhibited wind noise amplitudes of over $1\text{e-}07$ m/m/s in the 0-20 Hz frequency band, strong enough to mask signals from moderate earthquakes. We illustrated the strong impact of wind noise by comparing recordings of a simulated earthquake with and without artificial wind. A signal that was detectable on all unburied fiber segments in the absence of wind became undetectable on OD0.9 and OD3 unburied in the presence of wind. This suggests that wind is likely a major contributor to the poor signal quality often observed in unburied deployments on Earth, consistent with other studies that have identified wind as a dominant noise source in surface DAS deployments (Carr et al., 2025; Hudson et al., 2021; Viens et al., 2025).

While OD5.5 is representative of the fiber types commonly used in terrestrial DAS applications, we included the much lighter OD0.9 cable in our study to explore options more suitable for lunar deployment, where minimizing weight is a critical factor. Although cables like OD0.9 would not be typically used in terrestrial surface DAS applications, understanding its coupling behavior helps assess the feasibility of lightweight DAS systems for future lunar seismic monitoring. The experimental results for OD5.5 provide insight into the dominant factors affecting signal quality in unburied terrestrial deployments. This cable maintained stable amplitude and phase responses even when unburied throughout the tested frequency range, with only minor degradation compared to its shallow-buried configuration. Wind did lead to a significant increase of the noise floor, but the cable remained capable of detecting strong seismic events. In contrast, surface undulations had a strong detrimental impact: already relatively short (15-25 cm) free-hanging segments led to significant amplitude reduction and erratic phase behavior, particularly at higher frequencies. These results suggest that unburied DAS can achieve signal quality comparable to buried configurations in the absence of strong wind and if continuous ground contact is maintained. The feasibility of this depends on the deployment environment, but strategies such as pressing the cable down or flattening the surface could be employed to increase cable-ground contact.

4.2 Coupling mechanisms

Our experimental results show that signal quality in unburied fiber deployments varies strongly with cable type and deployment details. Thinner cables and configurations with reduced ground contact consistently yielded lower amplitudes and less reliable phase information. While these empirical patterns are clear, the physical mechanisms that govern strain transfer from the ground to the optical fiber remain poorly understood. In this section, we discuss possible coupling mechanisms and present a hypothesis that may explain the observed variability in signal quality. We begin by outlining the two-step process of strain transfer from the ground to the fiber and then explore how deployment geometry and cable properties influence each step. Strain transfer from the ground to the optical fiber involves two steps: 1) Ground-to-cable-coupling: Transfer of strain from the ground to the cable's outer surface; 2) Cable-to-fiber transfer: Transmission of strain through the cable layers to the optical fiber. The second step depends only on the cable's internal mechanical properties. Reinsch et al. (2017) developed a theoretical model for strain transfer in multilayer fiber-optic cables, showing that the efficiency of strain transmission from the cable surface to the optical fiber depends on cable geometry and material properties, and varies with wavelength. Their analysis highlights that strain transfer improves with increasing wavelength and is nearly ideal for typical seismic wavelengths. The first step, which we refer to as cable-ground coupling, depends on the contact points between cable and ground, the cable geometry in between contacts, both depending on the properties of the cable and the ground surface. In buried configurations with tight ground contact and no slip, coupling is assumed to be ideal—the cable moves with the ground. The no-slip assumption is discussed below. In that case, the strain transfer from the ground to the buried cable is purely determined

by the cable-to-fiber transfer. In unburied setups, however, the cable only follows ground motion at contact points, but not necessarily on the segments in between. Understanding coupling mechanisms is crucial for generalizing results, predicting performance under lunar conditions (e.g., reduced gravity), and designing optimized cables. We hypothesize that the main coupling mechanism is bending stress relief described in the following. In our unburied deployment in the LRS the cables are draped on the surface as flat as possible, still there are usually a few mm to cm between individual contact points with the ground. In between those contact points the cables are suspended in the air due to their stiffness and slightly arching towards the ground due to gravity. When the ground deforms and neighboring contact points move toward each other, the cable segment suspended between them can bend perpendicularly to the strain direction (as illustrated in Figure 6). This bending allows the cable to relieve part of the stress rather than transferring it as axial strain to the fiber core. While a full quantification of this effect is beyond the scope of this study, beam theory suggests that the resistance to bending increases with both the Young's modulus and the fourth power of the cable diameter. This implies that thinner and more flexible cables bend more easily under horizontal shortening, thereby relieving more stress through bending and reducing the strain transferred to the fiber. This aligns with our experimental observations, where thinner cables consistently recorded lower amplitudes, indicating less efficient strain transfer. In addition to cable stiffness and diameter, the length of the suspended segment also plays a critical role: longer segments bend more easily under the same horizontal deformation, further reducing strain transfer. This aligns with our observation of decreased amplitudes for all cable types with increasing undulation length. When free-hanging segments become too long, resonance effects may dominate, further degrading strain transfer and introducing phase shifts.

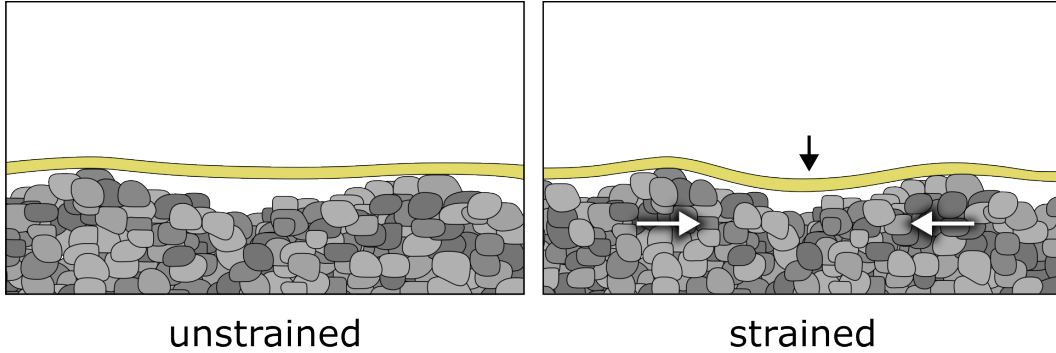


Figure 6. Conceptual illustration of the bending stress relief mechanism proposed to explain reduced strain transfer in unburied DAS cables. When ground deformation causes contact points to move closer or further apart, suspended cable segments bend rather than compress or stretch, relieving stress and reducing axial strain transmission to the fiber core.

Slip between the cable and the ground is another potential mechanism affecting coupling. Slip occurs when the force required to deform the cable exceeds the frictional force at the contact points. If slip were to occur, it would be amplitude-dependent: small deformations could be sustained by friction, while larger deformations might exceed the frictional threshold, leading to slip. This would imply a non-linear coupling behavior at the onset of slip. To assess whether slip is likely under realistic seismic conditions, we compared the frictional force at the cable-ground contact to the deformation force required to deform the cable. The frictional force is estimated as $F_f = \mu \cdot m_c \cdot g \approx 0.07 \text{ N/m}$, where $\mu \approx 0.5$ is the friction coefficient which can vary depending on the surface properties, $m_c = 0.015 \text{ kg/m}$ is an average cable mass per unit length, and g is gravitational acceleration. The deformation force, calculated using Hooke's law and assuming strain

amplitudes typical of seismic waves, is given by $F_d = \frac{2AE\varepsilon_{\max}}{\lambda} \approx 0.0012 - 0.012$ N for small strains. For stronger, active sources, this could increase to the range of 0.05–0.5 N. To summarize the slip condition, we define a dimensionless number comparing the available frictional force to the required deformation force:

$$\Gamma = \frac{F_f}{F_d} = \frac{\mu \cdot m_c \cdot g \cdot \lambda}{2AE\varepsilon_{\max}}, \quad (1)$$

where $\Gamma > 1$ indicates that friction is sufficient to prevent slip. For the parameter ranges considered, $\Gamma > 1$, suggesting that slip is unlikely under most realistic seismic scenarios. This conclusion is consistent with our experimental observations: we did not detect any non-linear behavior indicative of slip, even at amplitudes larger than those expected from natural seismic events. However, slip may still be relevant under conditions with low friction, such as due to substrate or cable properties, or under reduced lunar gravity, which lowers the normal force and thereby the available friction. Harmon et al. (2024) hypothesize that slip may have occurred in their unburied DAS experiment due to the low-friction properties of the cinder and ash substrates, and suggest that such slip can act like attenuation and cause frequency-dependent phase shifts.

4.3 Implications for lunar DAS

Our results indicate that unburied DAS may be viable for lunar applications, especially with robust cables like the OD5.5. Both OD5.5 and OD3 maintained relatively constant amplitude and phase response when unburied, suggesting reliable strain transfer in the tested frequency range. However, the detrimental effects of free-hanging segments underscore the importance of careful deployment. We recommend unspooling the cable onto the surface and pressing it down, e.g., by rolling a wheel over it, to ensure continuous contact. This simple method could significantly enhance coupling, as even minor deployment variations had substantial effects in our experiments. A similar concept has been proposed by Burns et al. (2023), who suggest rolling out a flat tape tether for lunar deployment. In our experiments, the thicker and stiffer OD5.5 consistently showed better coupling and higher recorded amplitudes. This observation is consistent with the predictions of our hypothesized coupling mechanisms. Our results also show that wind noise is a major factor degrading signal quality in unburied DAS on Earth, particularly for lighter cables like the OD0.9. This issue would not be present on the Moon due to the absence of an atmosphere, representing a clear advantage for lunar deployments. An important consideration for lunar deployment is the effect of reduced gravity on cable-ground coupling. We observed that heavier and stiffer cables, such as the OD5.5, exhibited better coupling quality. If cable weight is indeed a dominant factor in achieving good coupling, then replicating this behavior on the Moon, where gravity is approximately one-sixth that of Earth, might require cables that are six times heavier to exert the same normal force on the ground. However, it remains unclear whether weight alone governs coupling quality. Other factors such as cable diameter and stiffness may play equally or more significant roles, particularly if bending stress relief is indeed a relevant factor. Since these properties are independent of gravity, they would remain unchanged on the Moon. This uncertainty highlights the need for a deeper understanding of the physical mechanisms behind strain transfer.

Open Research Section

The used data sets and data processing codes are openly available at the ETH Zurich Research Collection via <https://doi.org/10.3929/ethz-c-000785214> (Probst et al., 2025).

Conflict of Interest disclosure

The authors declare there are no conflicts of interest for this manuscript.

Acknowledgments

SP and JOAR acknowledge funding from the Swiss National Science Foundation, grant 200021_212064/1. CD was supported by the Los Alamos National Laboratory (LANL) through its Center for Space and Earth Science (CSES). CSES is funded by LANL's Laboratory Directed Research and Development (LDRD) program under project number 20240477CR. AZ acknowledges support from: LANL Directed Research and Center for Space and Earth Sciences 20240065DR/YW0300. We thank LASN, Peter Roberts, and Emily Rodriguez for the use of seismometers and for helpful insight, and William Redford for measurements of wind speed. This manuscript has been assigned the number LA-UR-25-29704 by Los Alamos National Laboratory.

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Supplementary Material

1 Non-Linearity

Figure 1 shows the average fiber response amplitude (averaged over all frequencies) as a function of the driving amplitude of the shaker. If the response were completely linear, then all data points would fall on the interpolated straight lines. The observed amplitudes match this linear assumption relatively well. A few fiber segments deviate from the linear assumption (OD0.9 shallow buried, OD3 shallow buried, OD0.9 unburied). This is likely due to a slight drift of the channel locations during the acquisition caused by temperature changes of the interrogator. A correction was applied but some drift might remain, leading to a small error in the acquired amplitudes.

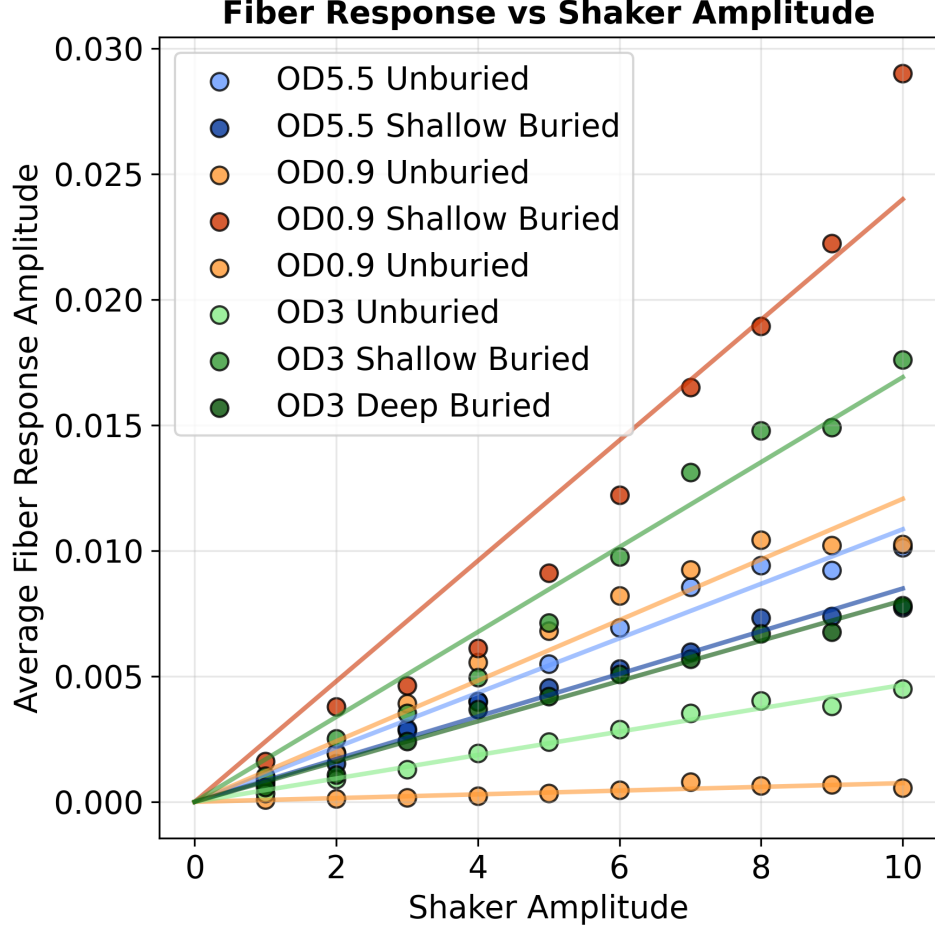


Figure 1: Average fiber response amplitude vs. shaker driving amplitude. Linear interpolation lines are shown for reference.

2 Time-variability of phase-shift

To further investigate the phase behavior of fiber segments that exhibit deviations from the reference phase, we performed a time-resolved phase analysis. In the main analysis, phase shifts were computed as averages over the full 250-cycle excitation period. However, several of these segments also showed irregular amplitude responses at the same frequencies, raising the question of whether the observed phase shifts reflect stable physical phenomena or are artifacts of temporal averaging over a time-varying phase.

To address this, we segmented the excitation period into shorter time windows and computed amplitude and phase independently for each segment. An example of this analysis is shown in Figure 2, where amplitude and phase are plotted for 0.1-second windows during excitation at 23 Hz and amplitude level 8. The results demonstrate that both amplitude and phase remain stable throughout the excitation period, indicating that the observed phase shifts are not due to random fluctuations but represent consistent offsets. This behavior was confirmed across multiple fiber segments and deployment configurations. These findings support the interpretation that the phase shifts observed in certain unburied configurations are physically meaningful and likely related to resonance effects, rather than being artifacts of averaging over unstable signals.

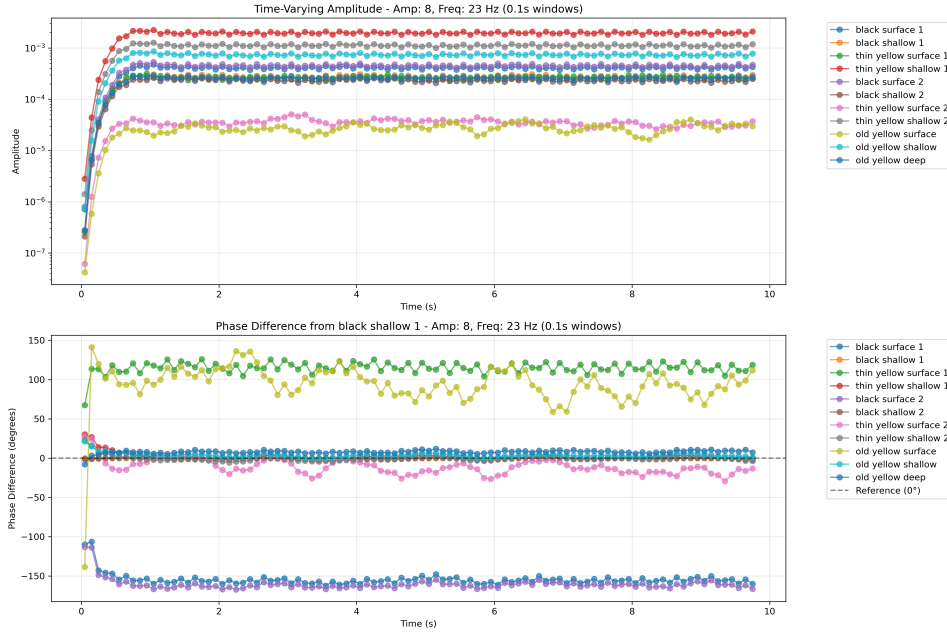


Figure 2: Amplitude and phase of all fiber segments for 0.1 second windows of the excited duration. Example shown for excitation at 23 Hz, at amplitude level 8.

3 Spectra of simulated earthquakes with and without wind

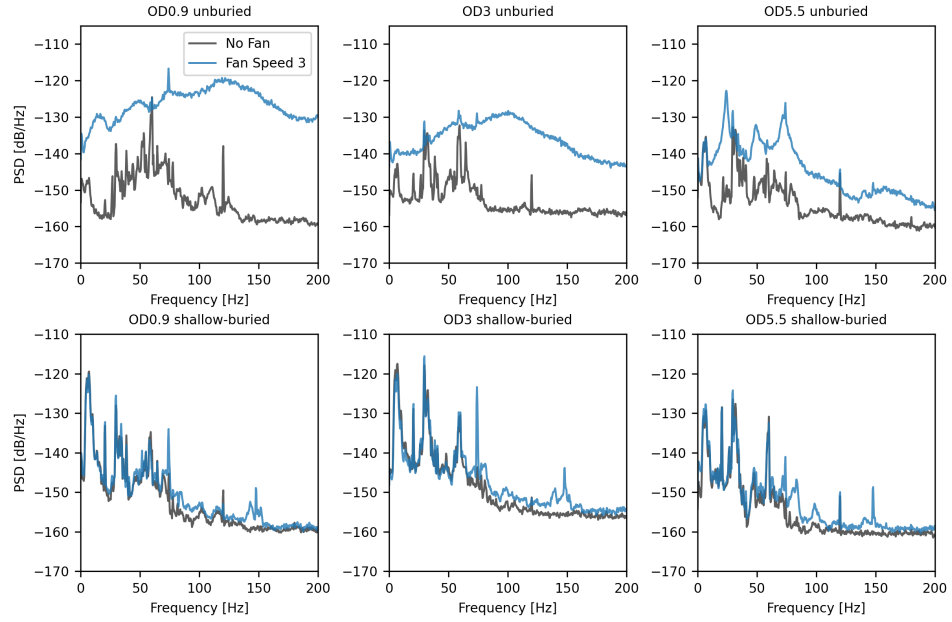


Figure 3: Power spectral density (PSD) comparison of simulated earthquake recordings with and without wind-induced noise across different cable segments. Top row: unburied cables (OD0.9, OD3, OD5.5); bottom row: corresponding shallow-buried segments. Each plot shows PSD curves from 0-200 Hz.

4 Absolute seismometer amplitudes

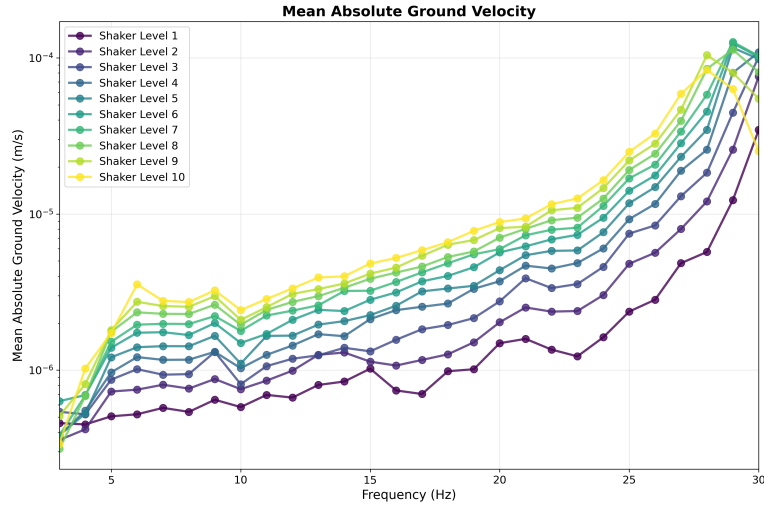


Figure 4: Mean absolute ground velocity measured by the seismometer during shaker tests across frequencies from 3 to 30 Hz. The data demonstrates how ground velocity varies with both frequency and excitation amplitude, with maximum velocities reaching approximately 10^{-4} m/s.